Comp9417 Week8

Ensemble Learning

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Disclaimer:

These slides are my own supplementary resource and are not official course materials. Any mistakes are my own.

Please refer to the solutions posted on **Moodle** as the official source.

Recap: AdaBoost

AdaBoost (Adaptive Boosting)

- $w_i = \frac{1}{m} \ \forall \ i$
- For j = 1 to k do
  - Learn L_j with data weight w
  - $\epsilon_j = \sum_{i=1}^{m} w_i^{(j)} 1[L_j(\mathrm{x}_i) \neq y_i] / \sum_{i=1}^{m} w_i^{(j)}$
  - $\lambda_j = \frac{1}{2} \log(\frac{1-\epsilon_j}{\epsilon_j})$
  - $w_i^{(j+1)} = w_i^{(j)} \exp(\lambda_j)$ for misclassified instances
  - $w_i^{(j+1)} = w_i^{(j)} \exp(-\lambda_j)$ for correct instances
- End

- Make prediction using the final model: $y(\mathrm{x}) = sign(\sum_{j=1}^{k} \lambda_i L_j(\mathrm{x}))$

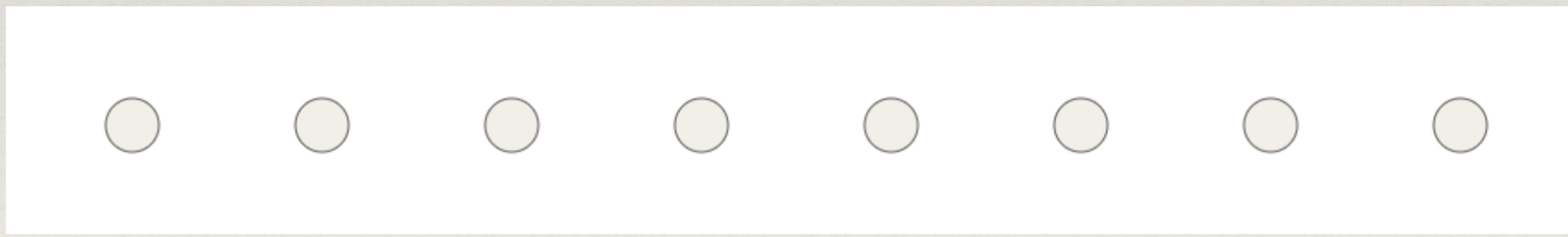
Recap: AdaBoost

Initialize the weight

$$w_i = \frac{1}{m} \quad \forall\, i$$

m: size of the data set

m=8



Recap: AdaBoost

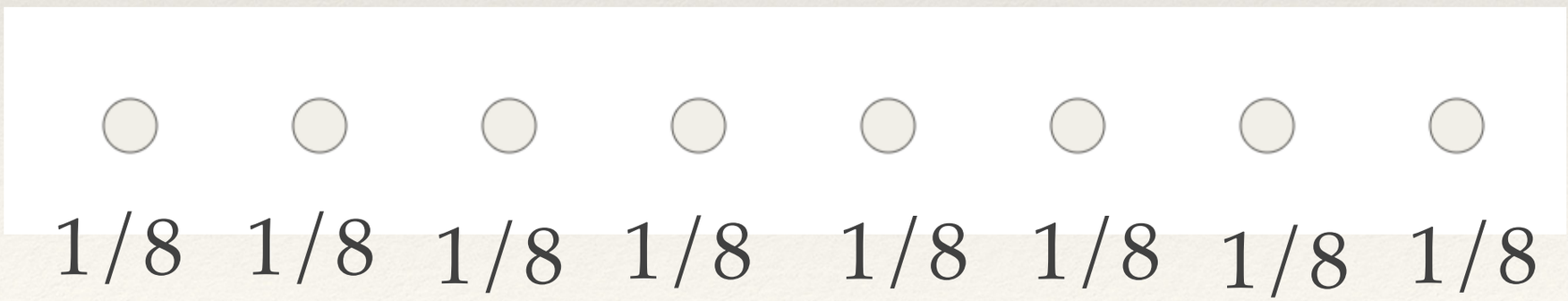
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m: size of the data set

m=8

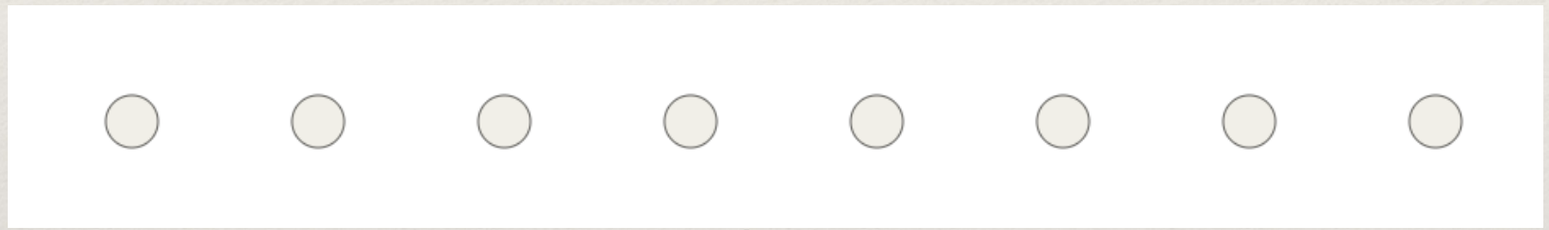
w_i = 1/8



Recap: AdaBoost

Training the Model

- $w_i = \frac{1}{m} \;\; \forall \, i$
- For j = 1 to k do
  - Learn L_j with data weight w



Learning

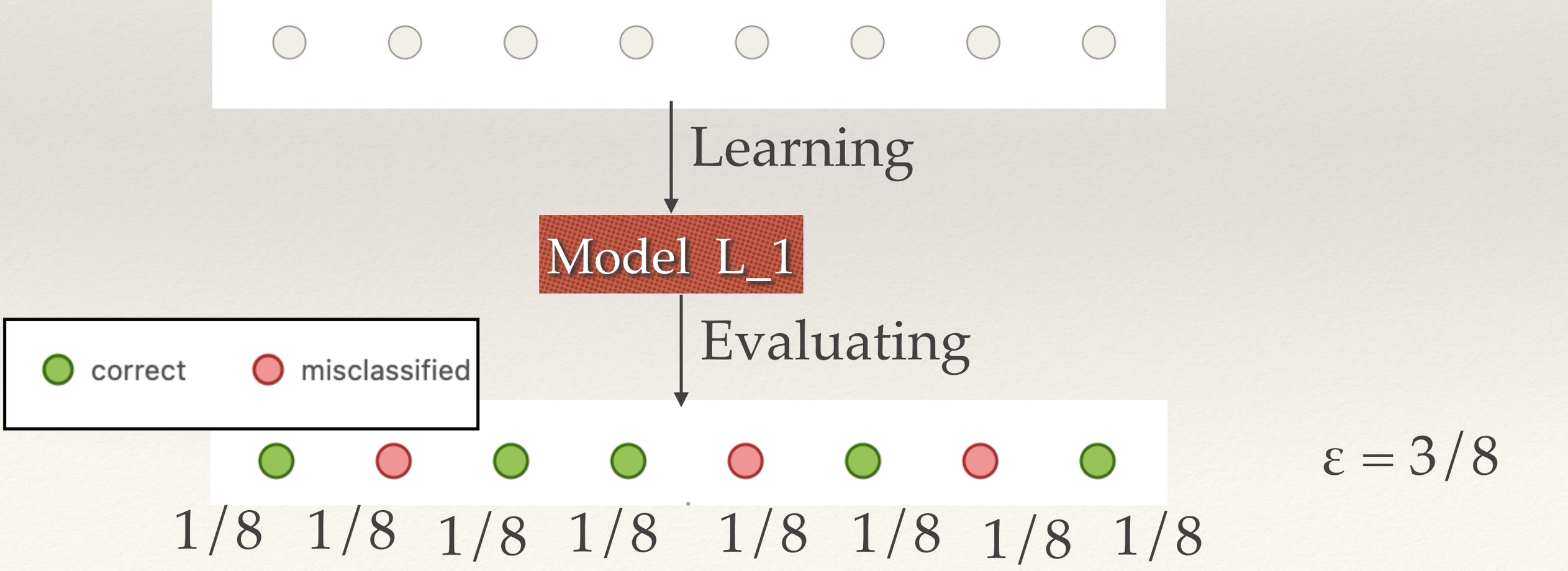
Model L_1

Recap: AdaBoost

Evaluating

- $w_i = \frac{1}{m} \;\; \forall \, i$
- For j = 1 to k do
  - Learn L_j with data weight w
  - $\epsilon_j = \sum_{i=1}^{m} w_i^{(j)} 1[L_j(\mathrm{x}_i) \neq y_i] / \sum_{i=1}^{m} w_i^{(j)}$

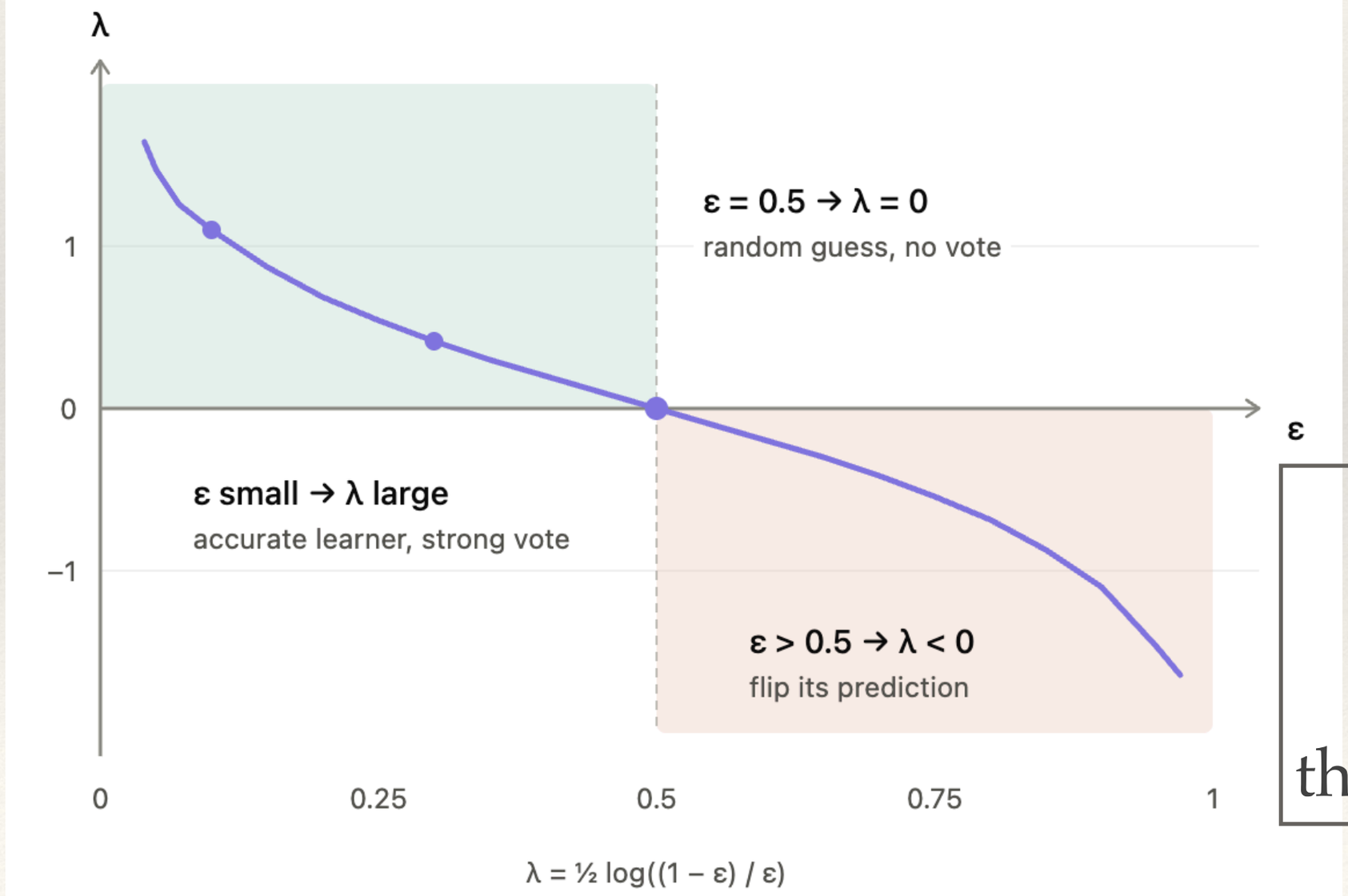
ε: Weighted Error



Evaluating

- $w_i = \frac{1}{m} \quad \forall\, i$
- For j = 1 to k do
  - Learn L_j with data weight w
  - $\epsilon_j = \sum_{i=1}^{m} w_i^{(j)} 1[L_j(\mathrm{x}_i) \neq y_i] / \sum_{i=1}^{m} w_i^{(j)}$
  - $\lambda_j = \frac{1}{2}\log(\frac{1-\epsilon_j}{\epsilon_j})$

λ: how much we trust this model

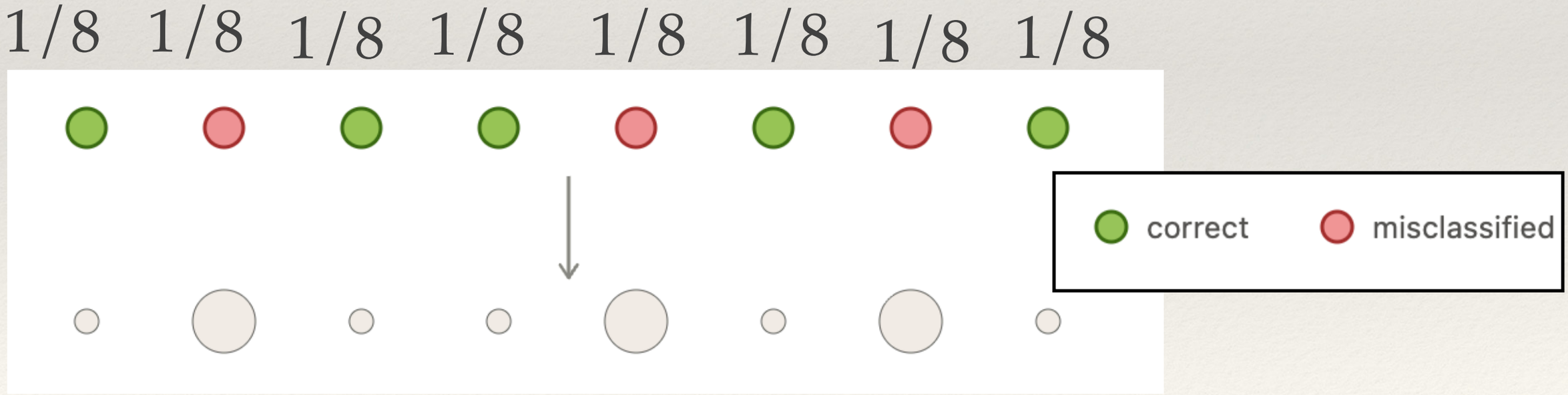


In general,, λ>0

Definition of Weak learner: does slightly better than random guessing.

Adjust the weights (of the instances).

- $w_i = \frac{1}{m} \ \forall \ i$
- For j = 1 to k do
  - Learn L_j with data weight w
  - $\epsilon_j = \sum_{i=1}^{m} w_i^{(j)} 1[L_j(\mathrm{x}_i) \neq y_i] / \sum_{i=1}^{m} w_i^{(j)}$
  - $\lambda_j = \frac{1}{2} \log(\frac{1-\epsilon_j}{\epsilon_j})$
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  - $w_i^{(j+1)} = w_i^{(j)} \exp(-\lambda_j)$ for correct instances



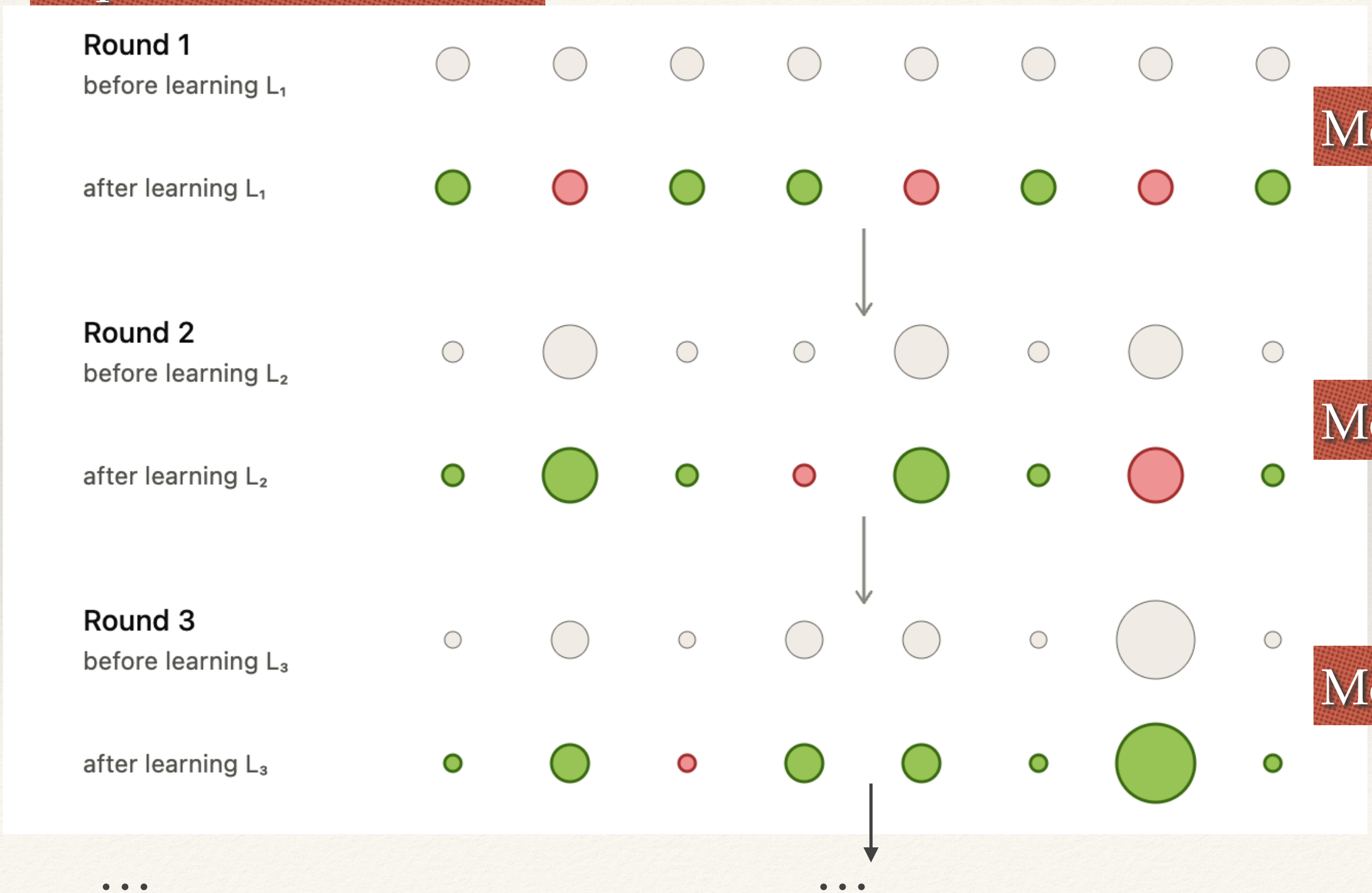
Recap: AdaBoost

Repeat this for k times

- $w_i = \frac{1}{m} \;\; \forall \, i$
- For j = 1 to k do
  - Learn L_j with data weight w
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- End

Recap: AdaBoost

Repeat this for k times



Model L_1

Model L_2

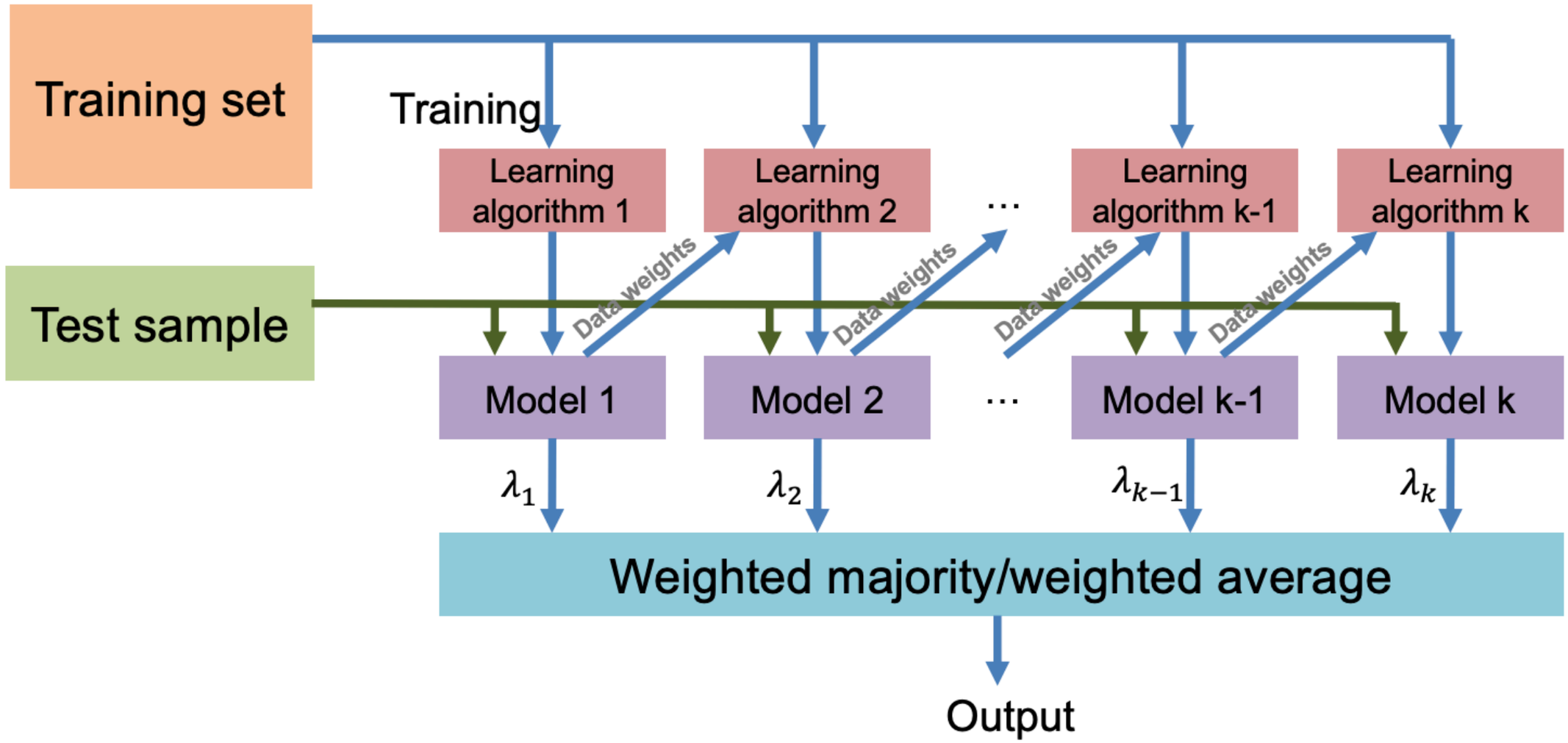
Model L_3

. . .

. . .

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Recap: AdaBoost



Question 1

Question 1 (Exponential Loss AdaBoost Derivation)

The goal of this question will be to understand one approach to deriving the AdaBoost algorithm. We have access to a data set:

$$\{(x_i, y_i) : i = 1, \ldots, n,\ x_i \in \mathcal{X},\ y_i \in \{-1, 1\}\},$$

the notation here implies that the features x_i take values in some set $\mathcal{X}$, for example $\mathcal{X} = \mathbb{R}^d$, and the labels y_i are either $+1$ or -1. Consider the class of weak learners:

$$\mathcal{H} = \{h : \mathcal{X} \to \{-1, 1\}\}.$$

The notation here means that each hypothesis h in $\mathcal{H}$ is a function taking values in $\mathcal{X}$ and returning values in $\{-1, 1\}$. Next, define the following exponential loss:

$$L(h, y) := L(h) = \sum_{i=1}^{n} \ell(h(x_i), y_i), \qquad \ell(h(x_i), y_i) = e^{-y_i h(x_i)}.$$

Q1 Analyze

Dataset $\{(x_i, y_i) : i = 1, \ldots, n,\ x_i \in \mathcal{X},\ y_i \in \{-1, 1\}\},$

Features x_i (d features) $\mathcal{X} = \mathbb{R}^d,$

Labels y_i $y_i \in \{-1, 1\}$

Models / hypothesis / weak learners: $\mathcal{H} = \{h : \mathcal{X} \to \{-1, 1\}\}.$

Loss Function: $L(h, y) := L(h) = \sum_{i=1}^{n} \ell(h(x_i), y_i), \qquad \ell(h(x_i), y_i) = e^{-y_i h(x_i)}.$

Q1a

(a) Describe the behaviour of the loss function ℓ. What happens when our hypothesis returns the correct label ($h(x_i) = y_i$)? What about when an incorrect label is returned?

$$L(h, y) := L(h) = \sum_{i=1}^{n} \ell(h(x_i), y_i), \qquad \ell(h(x_i), y_i) = e^{-y_i h(x_i)}.$$

$$y_i \in \{-1, 1\}$$

Q1a Analyze

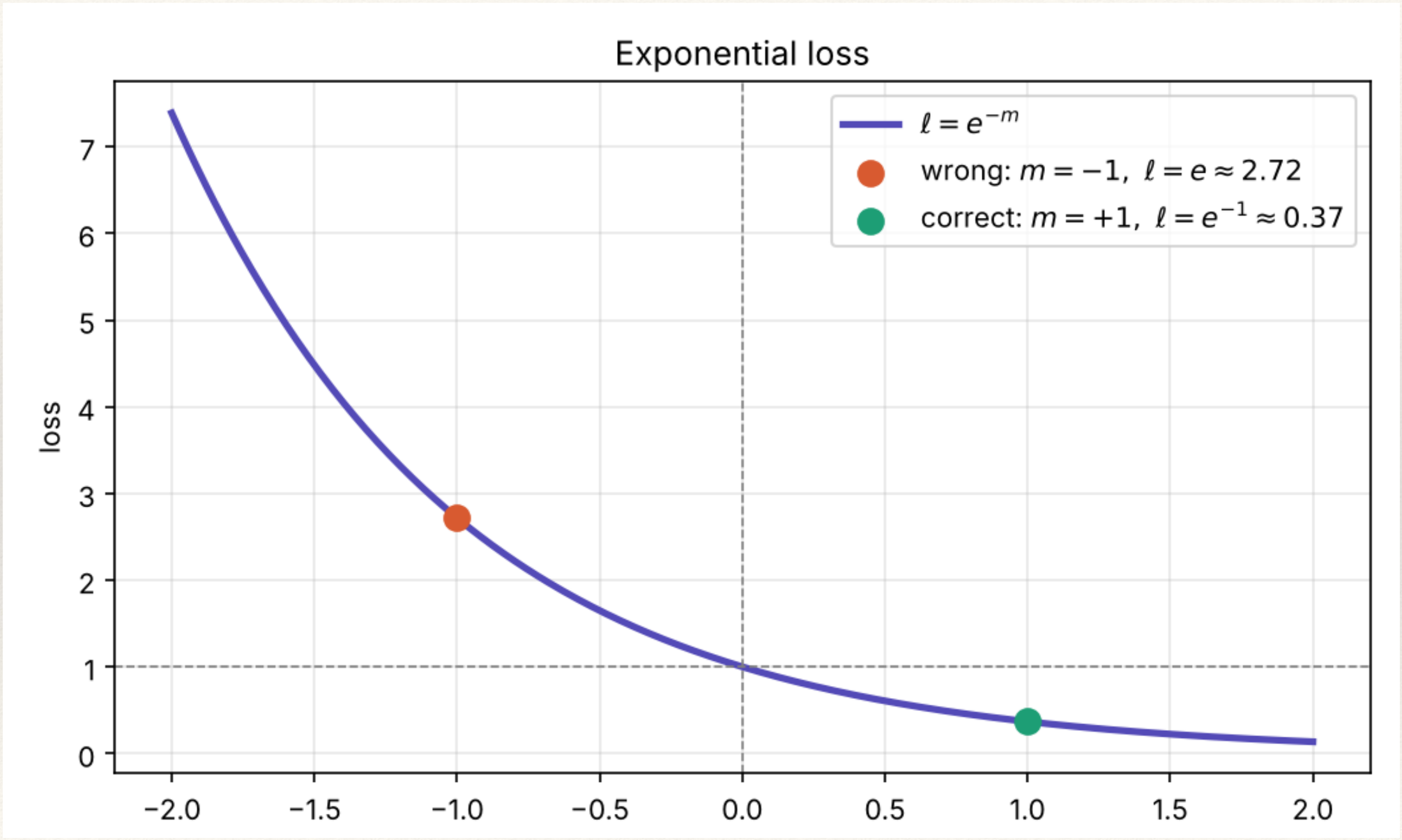
(a) Describe the behaviour of the loss function ℓ. What happens when our hypothesis returns the correct label ($h(x_i) = y_i$)? What about when an incorrect label is returned?

$$L(h, y) := L(h) = \sum_{i=1}^{n} \ell(h(x_i), y_i), \qquad \ell(h(x_i), y_i) = e^{-y_i h(x_i)}.$$

$$y_i \in \{-1, 1\}$$

y_i	$h(x_i)$	$y_i h(x_i)$	Outcome	Loss $e^{-y_i h(x_i)}$
+1	+1	+1	correct	$e^{-1} \approx 0.37$
+1	−1	−1	wrong	$e^{1} \approx 2.72$
−1	−1	+1	correct	$e^{-1} \approx 0.37$
−1	+1	−1	wrong	$e^{1} \approx 2.72$

Q1a Analyze


Exponential loss
$\ell = e^{-m}$
wrong: $m = -1,\ \ell = e \approx 2.72$
correct: $m = +1,\ \ell = e^{-1} \approx 0.37$
loss
0
1
2
3
4
5
6
7
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0

Q1b

(b) Recall that the AdaBoost algorithm seeks to recursively construct an ensemble classifier which at the $(m-1)^{\text{st}}$ step has the form:

$$\text{sign}(C_{m-1}(x)) = \begin{cases} +1 & \text{if } C_{m-1}(x) \geq 0 \\ -1 & \text{if } C_{m-1}(x) < 0, \end{cases}$$

where

$$C_{m-1}(x) = \alpha_1 h_1(x) + \alpha_2 h_2(x) + \cdots + \alpha_{m-1} h_{m-1}(x),$$

and the weights $\alpha_1, \ldots, \alpha_{m-1}$ are all positive. Assume that you have access to C_{m-1}, so that you know $\alpha_1, \ldots, \alpha_{m-1}$ and $h_1, \ldots, h_{m-1}$. The two challenges at this stage are then:

- How do we pick a new hypothesis h at step m?
- How do we pick the corresponding weight α_m?

Show that

$$L(C_m) = \sum_{i=1}^{n} w_i^{(m)} e^{-\alpha_m y_i h_m(x_i)}$$

where

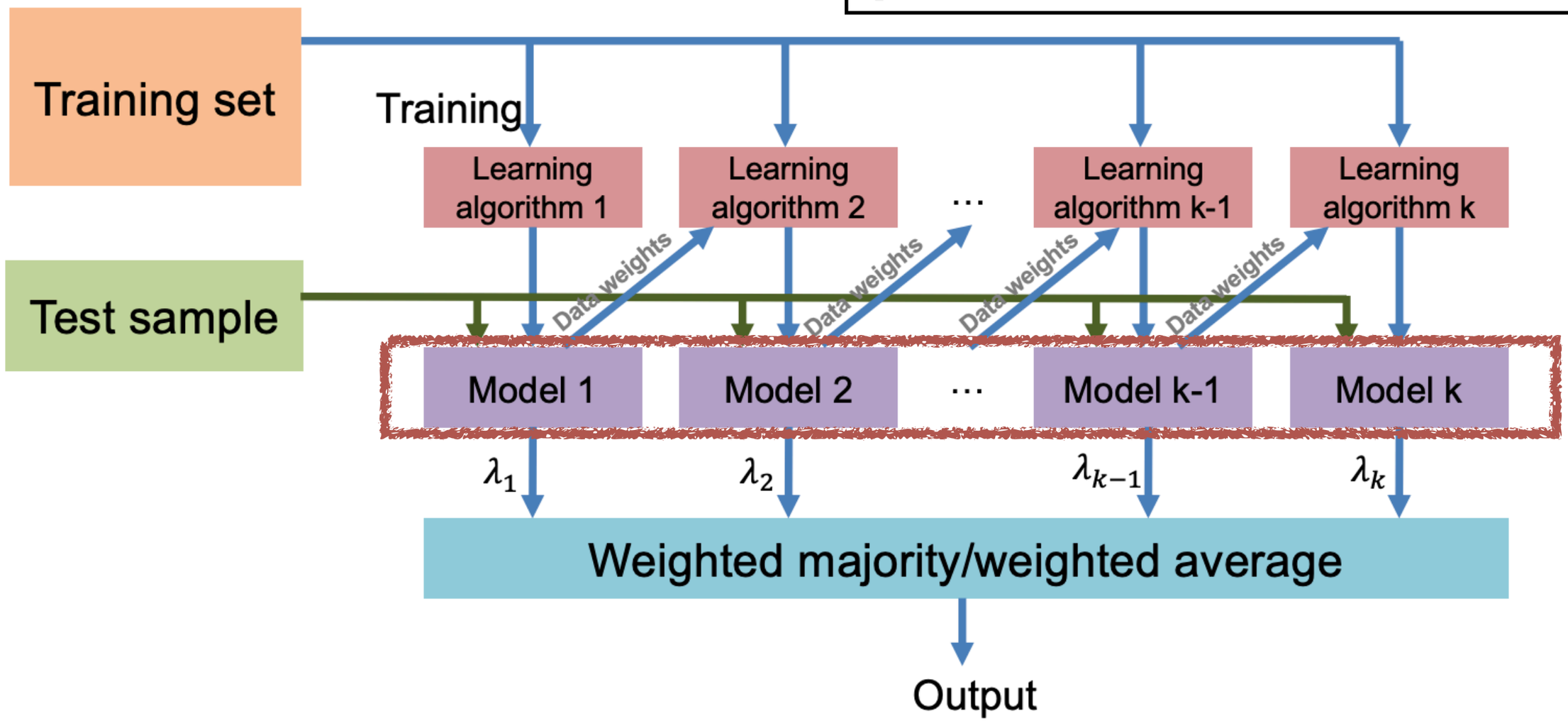
$$w_i^{(m)} := e^{-y_i C_{m-1}(x_i)}.$$

Explain why these weights $w_i^{(m)}$ for $i = 1, \ldots, n$ are larger for data points on which C_{m-1} made a mistake, and smaller for data points that were correctly classified by C_{m-1}.

Q1b Analyze

- How do we pick a new hypothesis h at step m?
- How do we pick the corresponding weight α_m?

> Notation: $\alpha_j \equiv \lambda_j$, $h_j \equiv L_j$ (as in lecture slides).



Q1b Analyze

Weighted Voting

$$C_{m-1}(x) = \alpha_1 h_1(x) + \alpha_2 h_2(x) + \cdots + \alpha_{m-1} h_{m-1}(x),$$

Show that

$$L(C_m) = \sum_{i=1}^{n} w_i^{(m)} e^{-\alpha_m y_i h_m(x_i)}$$

where

$$w_i^{(m)} := e^{-y_i C_{m-1}(x_i)}.$$

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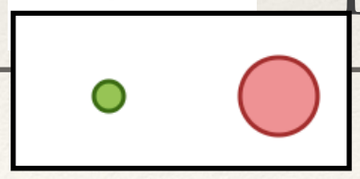
Q1b Analyze

Weighted Voting

$$L(C_m) = \sum_{i=1}^{n} w_i^{(m)} e^{-\alpha_m y_i h_m(x_i)}$$

$$w_i^{(m)} := e^{-y_i C_{m-1}(x_i)}.$$

$L(C_m)$	Loss of the ensemble Model.
$\sum_{i=1}^{n}$	For each point … Then add values up.
$w_i^{(m)}$	weight/size of instance i at step (m).
$e^{-\alpha_m y_i h_m(x_i)}$	Loss of new model (m) at this point



Want to prove: Loss(Ensemble) = ∑ weight * Loss(new model_m)

Q1b Analyze. We already know:

Loss Function:

$$L(h, y) := L(h) = \sum_{i=1}^{n} \ell(h(x_i), y_i), \qquad \ell(h(x_i), y_i) = e^{-y_i h(x_i)}.$$

Weighted Voting:

$$C_{m-1}(x) = \alpha_1 h_1(x) + \alpha_2 h_2(x) + \cdots + \alpha_{m-1} h_{m-1}(x),$$

$$C_m(x) = \alpha_1 h_1(x) + \alpha_2 h_2(x) + \cdots + \alpha_{m-1} h_{m-1}(x) + \alpha_m h_m(x)$$



$$C_m(x) = C_{m-1}(x) + \alpha_m h_m(x)$$

Hint: (product rule for exponents)

$$e^{a+b} = e^a \cdot e^b$$

Q1b Answer

$$
\begin{aligned}
L(C_m) &= \sum_{i=1}^{n} e^{-y_i C_m(x_i)} \\
&= \sum_{i=1}^{n} e^{-y_i (C_{m-1}(x) + \alpha_m h_m(x_i))} \\
&= \sum_{i=1}^{n} e^{-y_i C_{m-1}(x_i)} e^{-\alpha_m y_i h_m(x_i)} \\
&= \sum_{i=1}^{n} w_i^{(m)} e^{-\alpha_m y_i h_m(x_i)}.
\end{aligned}
$$

$$
w_i^{(m)} := e^{-y_i C_{m-1}(x_i)}.
$$

Q1b Intuition

Loss:

$$\ell(h(x_i), y_i) = e^{-y_i h(x_i)}.$$

Weight:

$$w_i^{(m)} := e^{-y_i C_{m-1}(x_i)}.$$

C_(m-1) : the emsemble model at step (m-1).

The weight is just the loss of the previous ensemble on that point.

Q1c

(c) Show further that

$$L(C_m) = e^{-\alpha_m} \sum_{i:h_m(x_i)=y_i} w_i^{(m)} + e^{\alpha_m} \sum_{i:h_m(x_i)\neq y_i} w_i^{(m)}.$$

Note that the first sum is only summing over indices where the base learner h_m does not make a mistake, and the second sum is summing over indices on which a mistake is made. We can interpret the first sum as the component of the error due to making correct classification, and the second sum as the component due to misclassification.

Q1c Analyze

(c) Show further that

$$L(C_m) = e^{-\alpha_m} \sum_{i:h_m(x_i)=y_i} w_i^{(m)} + e^{\alpha_m} \sum_{i:h_m(x_i)\neq y_i} w_i^{(m)}.$$

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$$L(C_m) = \sum_{i=1}^{n} w_i^{(m)} e^{-\alpha_m y_i h_m(x_i)}$$

Q1c Analyze

(c) Show further that

$$L(C_m) = e^{-\alpha_m} \sum_{i:h_m(x_i)=y_i} w_i^{(m)} + e^{\alpha_m} \sum_{i:h_m(x_i)\neq y_i} w_i^{(m)}.$$

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$$L(C_m) = \sum_{i=1}^{n} w_i^{(m)} e^{-\alpha_m y_i h_m(x_i)}$$

Loss(Ensemble) = ∑ size * Loss(new model_m)



Q1c Analyze



$$L(C_m) = \sum_{i=1}^{n} w_i^{(m)} e^{-\alpha_m y_i h_m(x_i)}$$

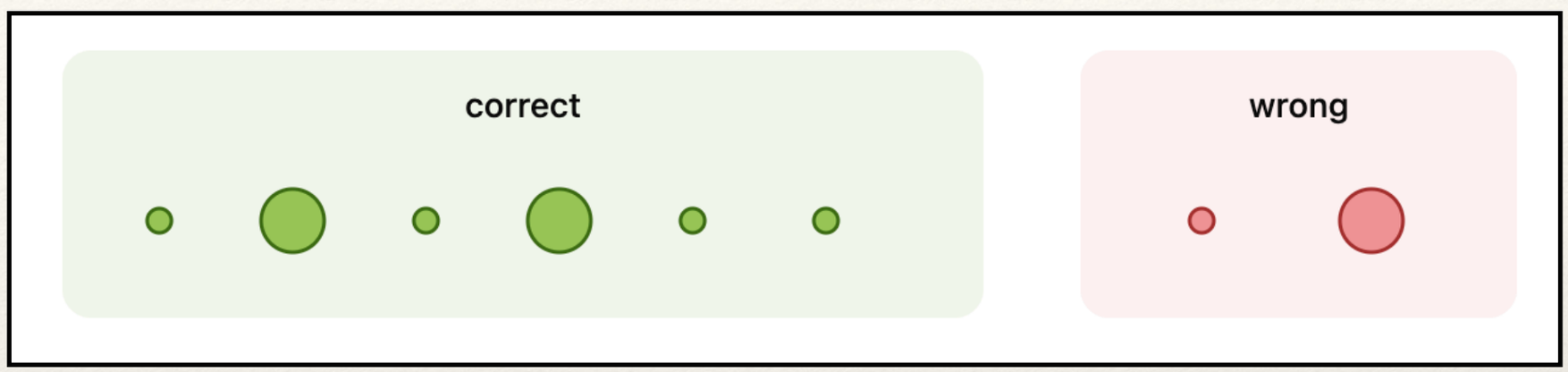
$$= \sum_{i:h_m(x_i)=y_i} w_i^{(m)} e^{-\alpha_m y_i h_m(x_i)} + \sum_{i:h_m(x_i)\neq y_i} w_i^{(m)} e^{-\alpha_m y_i h_m(x_i)}$$



Loss(Ensemble) = ∑ size * Loss(new model_m)

Q1c Analyze

Weight $w_i^{(m)}$: size of the point



$$
\begin{aligned}
&= \sum_{i:h_m(x_i)=y_i} w_i^{(m)} e^{-\alpha_m y_i h_m(x_i)} + \sum_{i:h_m(x_i)\neq y_i} w_i^{(m)} e^{-\alpha_m y_i h_m(x_i)} \\
&= \sum_{i:h_m(x_i)=y_i} w_i^{(m)} e^{-\alpha_m} + \sum_{i:h_m(x_i)\neq y_i} w_i^{(m)} e^{\alpha_m}
\end{aligned}
$$

Loss: $e^{-\alpha_m y_i h_m(x_i)}$

y_i	$h(x_i)$	$y_i h(x_i)$	Outcome
+1	+1	+1	correct
+1	−1	−1	wrong
−1	−1	+1	correct
−1	+1	−1	wrong

Q1c Analyze

$$L(C_m) = \sum_{i=1}^{n} w_i^{(m)} e^{-\alpha_m y_i h_m(x_i)}$$

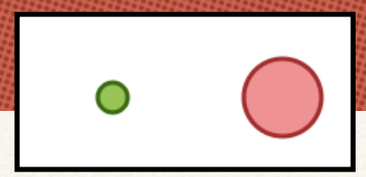
$$= \sum_{i:h_m(x_i)=y_i} w_i^{(m)} e^{-\alpha_m y_i h_m(x_i)} + \sum_{i:h_m(x_i)\neq y_i} w_i^{(m)} e^{-\alpha_m y_i h_m(x_i)}$$

$$= \sum_{i:h_m(x_i)=y_i} w_i^{(m)} e^{-\alpha_m} + \sum_{i:h_m(x_i)\neq y_i} w_i^{(m)} e^{\alpha_m}$$

$$= e^{-\alpha_m} \sum_{i:h_m(x_i)=y_i} w_i^{(m)} + e^{\alpha_m} \sum_{i:h_m(x_i)\neq y_i} w_i^{(m)}.$$

alpha_m: how much we trust this model

Loss(Ensemble) = ∑ size * Loss(new model_m)
= WeightedSum(size)



Q1d

(d) Since we want to find α_m, h_m, it makes sense to minimize this loss with respect to these parameters, that is, we want to find the base learner and weight that solve

$$\min_{\alpha_m > 0, h_m \in \mathcal{H}} L(C_m).$$

We want to minimize Loss.

Note that we only want to allow positive weights, and only hypotheses that belong to our set of weak learners. Argue heuristically that in order to minimize the loss, we should always pick h_m to minimize the (weighted) classification error, i.e. the best possible $h_m \in \mathcal{H}$ satisfies

$$h_m^* = \arg\min_{h_m \in \mathcal{H}} \sum_{i: h_m(x_i) \neq y_i} w_i^{(m)}.$$

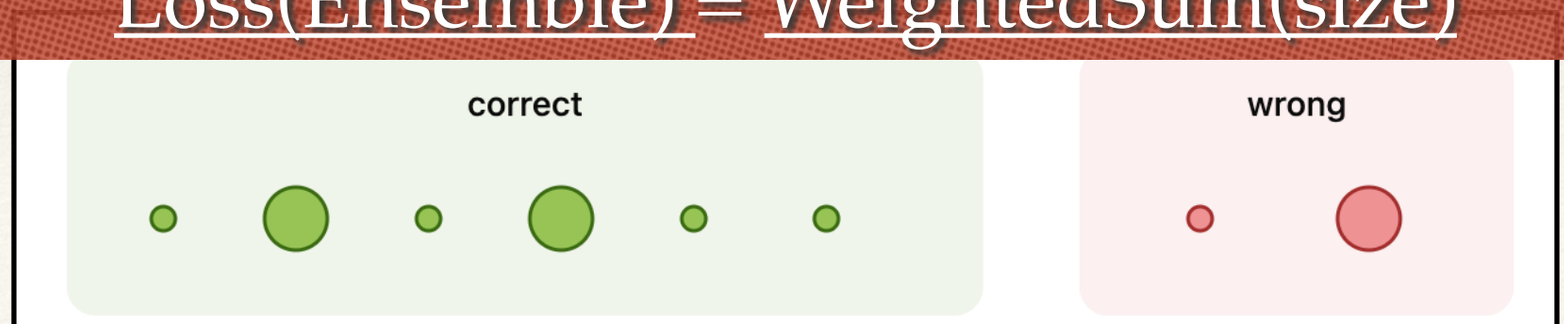
Just need to minimize this.

Explain why this choice of base learner can be interpreted as: trying to learn the mistakes of the previous learners.

Loss:

$$e^{-\alpha_m} \sum_{i: h_m(x_i) = y_i} w_i^{(m)} + e^{\alpha_m} \sum_{i: h_m(x_i) \neq y_i} w_i^{(m)}.$$

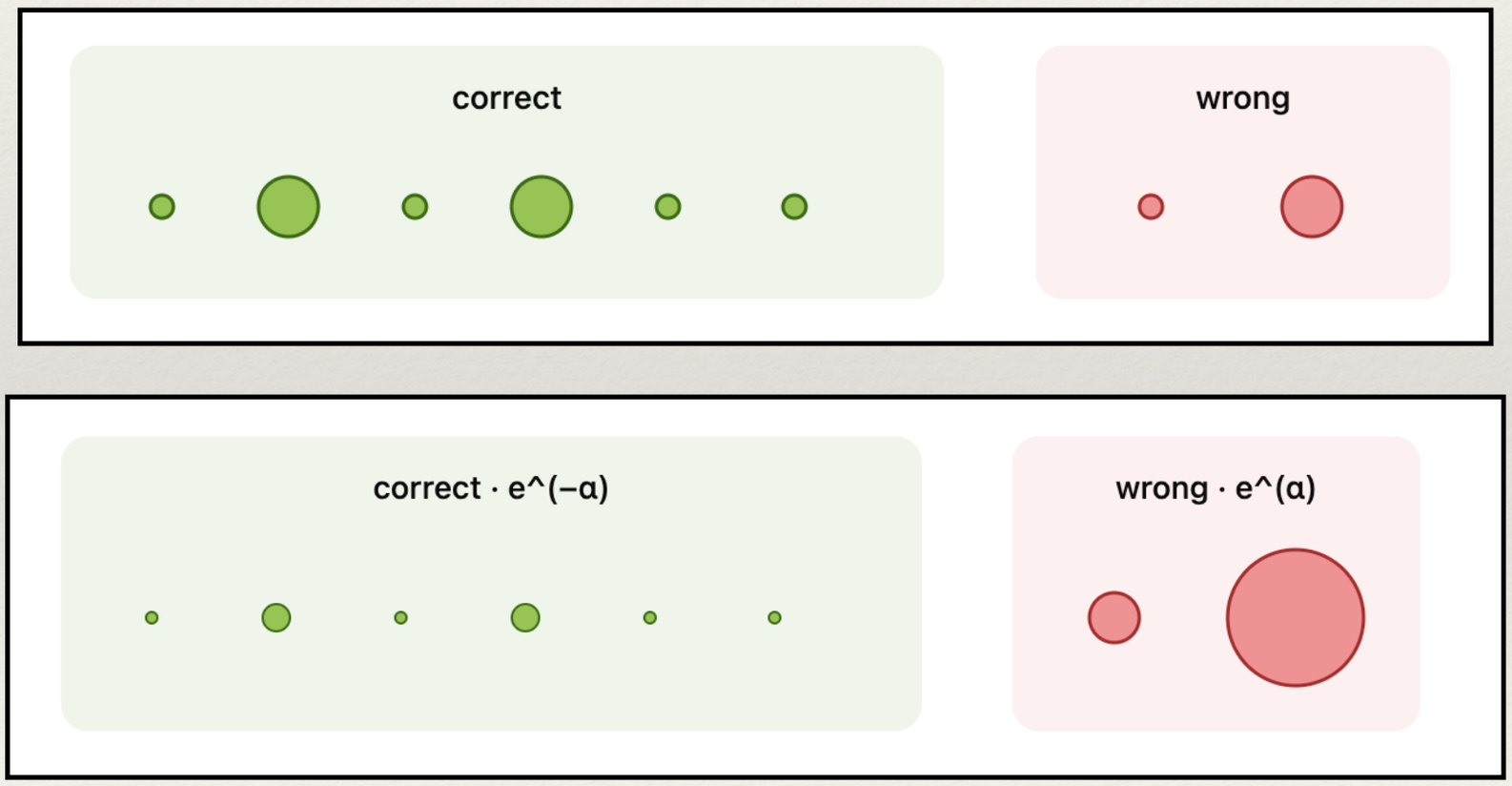
Loss(Ensemble) = WeightedSum(size)



Q1d Intuition

Loss:

$$e^{-\alpha_m} \sum_{i:h_m(x_i)=y_i} w_i^{(m)} + e^{\alpha_m} \sum_{i:h_m(x_i)\neq y_i} w_i^{(m)}.$$



Correct: Scaling down

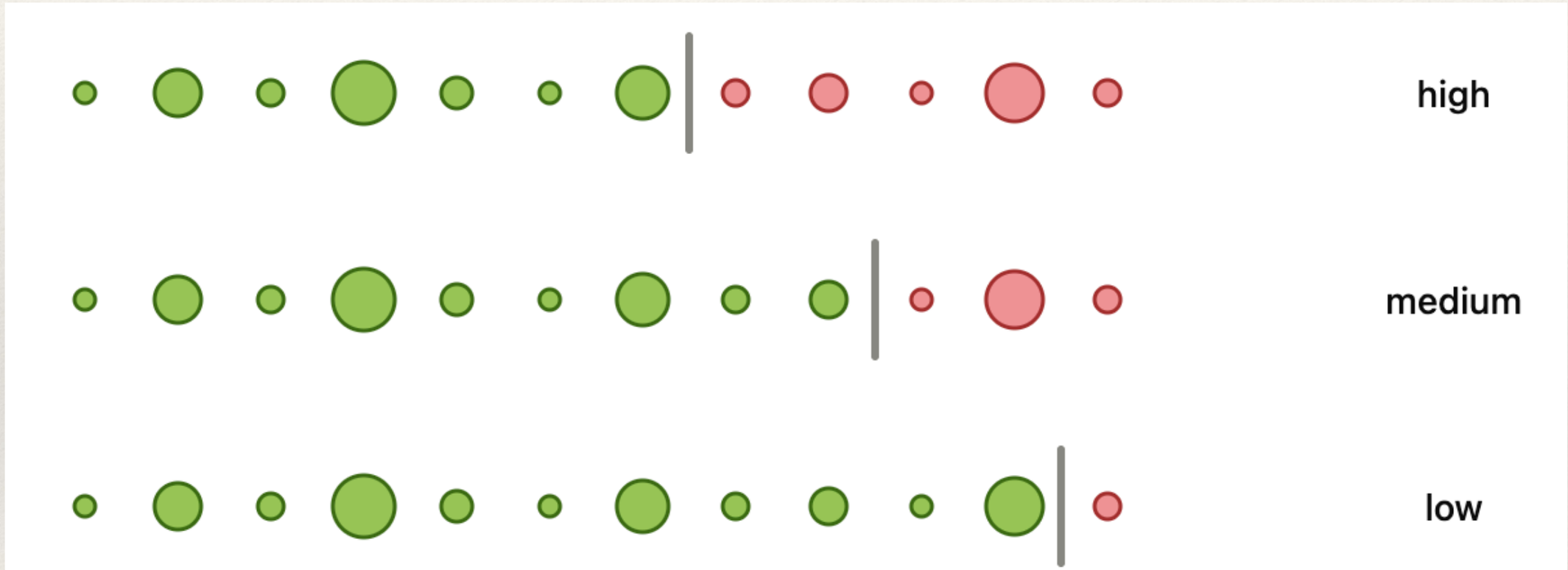
Wrong: Scaling up

Loss(Ensemble) = WeightedSum(size)

Q1d Analyze

The sum of the size is constant.

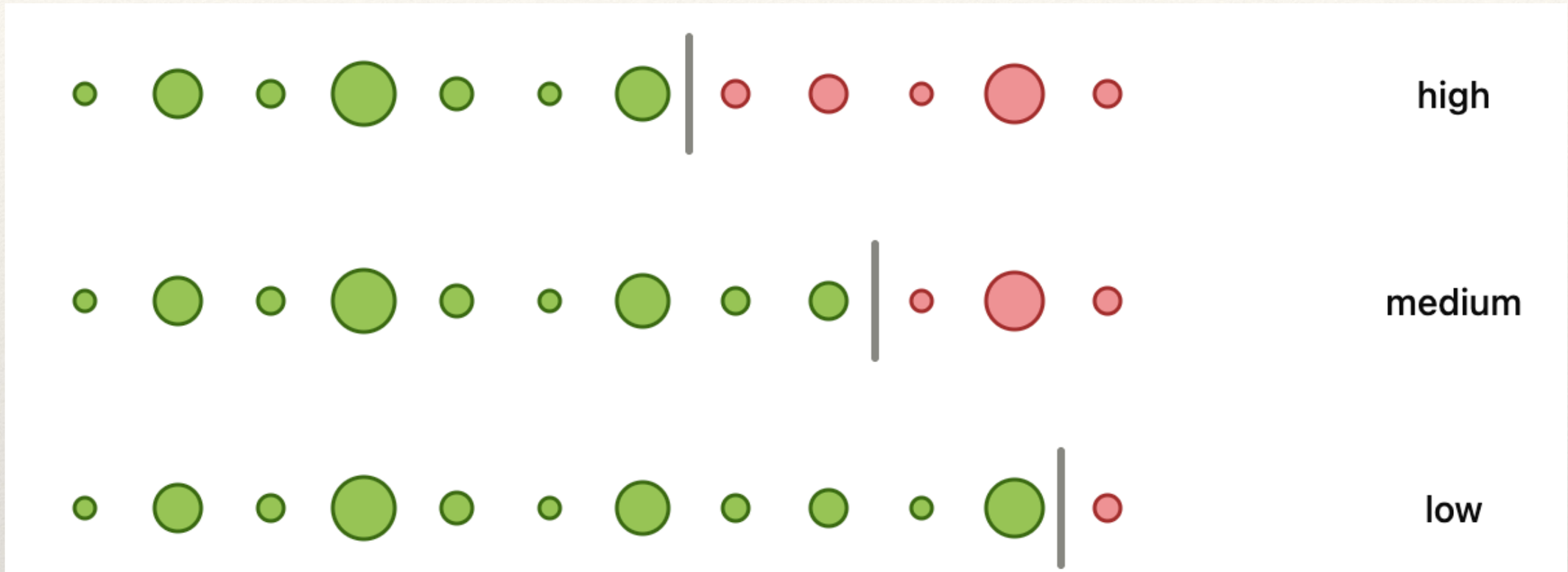
Correct: Scaling down Wrong: Scaling up



Minimise the Wrong part.

$$\sum_{i:h_m(x_i)\neq y_i} w_i^{(m)}.$$

Q1d Analyze



Minimise the wrong part.

What's More?

If must be Wrong, be Wrong on the small ones.

Explain why this choice of base learner can be interpreted as: trying to learn the mistakes of previous learners.

Q1e

(e) Suppose that we set $h_m = h_m^*$. What remains is to choose α_m. Solve the problem

$$\arg\min_{\alpha_m > 0} L(C_m^*, y) = \arg\min_{\alpha_m > 0} L(C_{m-1} + \alpha_m h_m^*, y)$$

by differentiating with respect to α_m and setting equal to zero. Show that the solution is

$$\alpha_m^* = \frac{1}{2} \ln \left(\frac{\sum_{i: h_m^*(x_i) = y_i} w_i^{(m)}}{\sum_{i: h_m^*(x_i) \neq y_i} w_i^{(m)}} \right).$$

Q1e Analyze

(e) Suppose that we set $h_m = h_m^*$. What remains is to choose α_m. Solve the problem

$$\arg\min_{\alpha_m > 0} L(C_m^*, y) = \arg\min_{\alpha_m > 0} L(C_{m-1} + \alpha_m h_m^*, y)$$

by differentiating with respect to α_m and setting equal to zero. Show that the solution is

$$\alpha_m^* = \frac{1}{2} \ln \left(\frac{\sum_{i:h_m^*(x_i)=y_i} w_i^{(m)}}{\sum_{i:h_m^*(x_i)\neq y_i} w_i^{(m)}} \right).$$

- $w_i = \frac{1}{m} \;\; \forall\, i$
- For j = 1 to k do
  - Learn L_j with data weight w
  - $\epsilon_j = \sum_{i=1}^{m} w_i^{(j)} 1[L_j(\mathrm{x}_i) \neq y_i] / \sum_{i=1}^{m} w_i^{(j)}$
  - $\lambda_j = \frac{1}{2}\log(\frac{1-\epsilon_j}{\epsilon_j})$

α: how much we trust this model

Q1e Analyze

Loss Function:

$$L(C_m) = e^{-\alpha_m} \sum_{i:h_m(x_i)=y_i} w_i^{(m)} + e^{\alpha_m} \sum_{i:h_m(x_i)\neq y_i} w_i^{(m)}.$$

We want to minimize it, so

$$0 = \frac{dL}{d\alpha_m}$$

$$\alpha_m^* = \frac{1}{2} \ln \left(\frac{\sum_{i:h_m^*(x_i)=y_i} w_i^{(m)}}{\sum_{i:h_m^*(x_i)\neq y_i} w_i^{(m)}} \right).$$

Q1f

(f) Define the weighted classification error proportion:

$$\epsilon_m := \frac{\sum_{i:h_m(x_i)\neq y_i} w_i^{(m)}}{\sum_{i=1}^{n} w_i^{(m)}}.$$

Show that

$$\alpha_m^* = \frac{1}{2}\ln\left(\frac{1-\epsilon_m}{\epsilon_m}\right).$$

$$\alpha_m^* = \frac{1}{2}\ln\left(\frac{\sum_{i:h_m^*(x_i)=y_i} w_i^{(m)}}{\sum_{i:h_m^*(x_i)\neq y_i} w_i^{(m)}}\right).$$

Explain why α_m^* is always positive, and further clarify the role played by α_m^*

α: how much we trust this model

Q1f Analyze

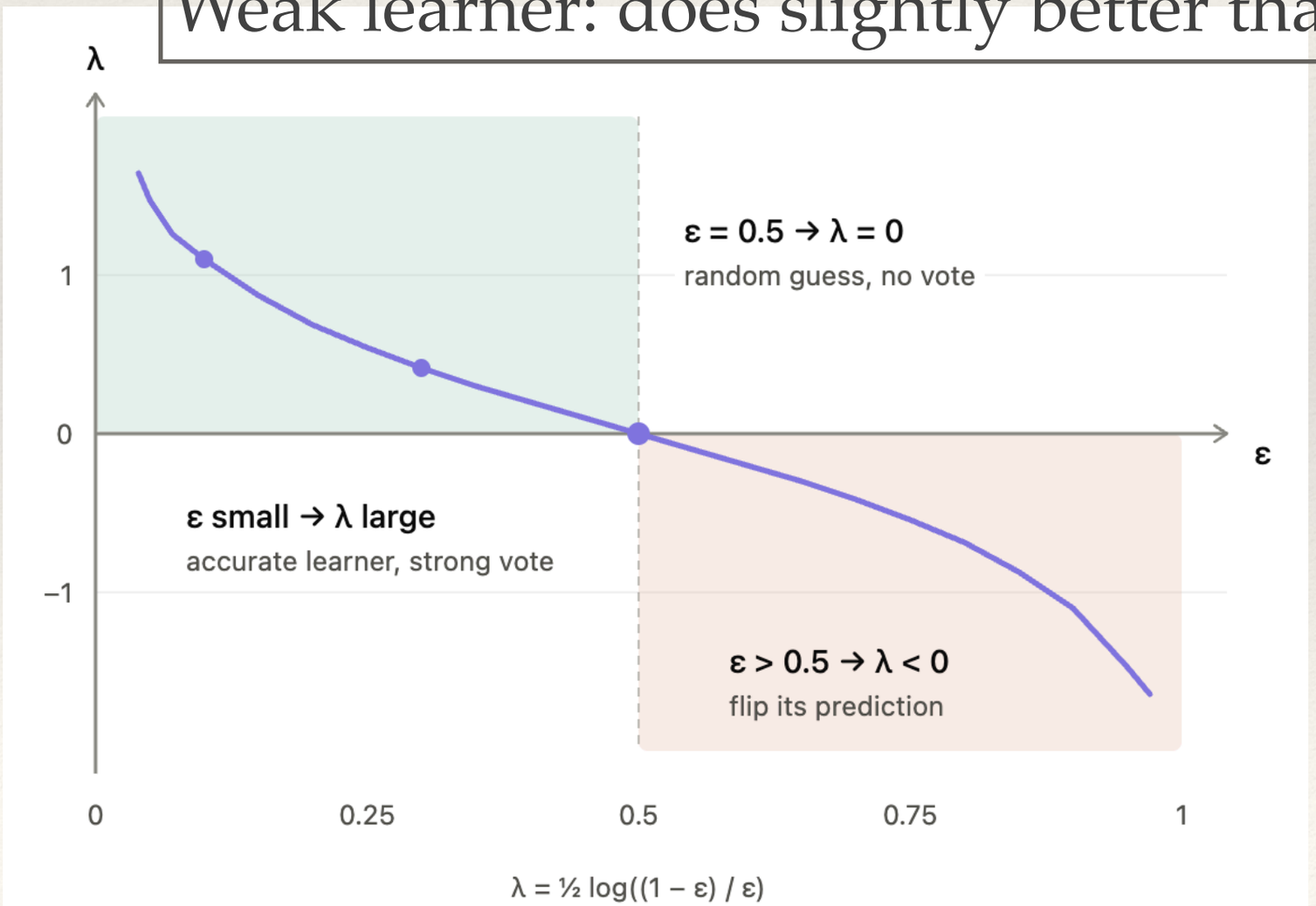
$$\alpha_m^* = \frac{1}{2}\ln\left(\frac{1-\epsilon_m}{\epsilon_m}\right).$$

$\frac{1-\epsilon_m}{\epsilon_m}$ Weighted accurate rate

Weighted error rate

Weak learner: does slightly better than random guessing.

$\epsilon_m = 0.5$



Q1g

(g) Finally, we need to formalize the change in weights at each iteration. Show that the weight updates can be written in the following recursive form:

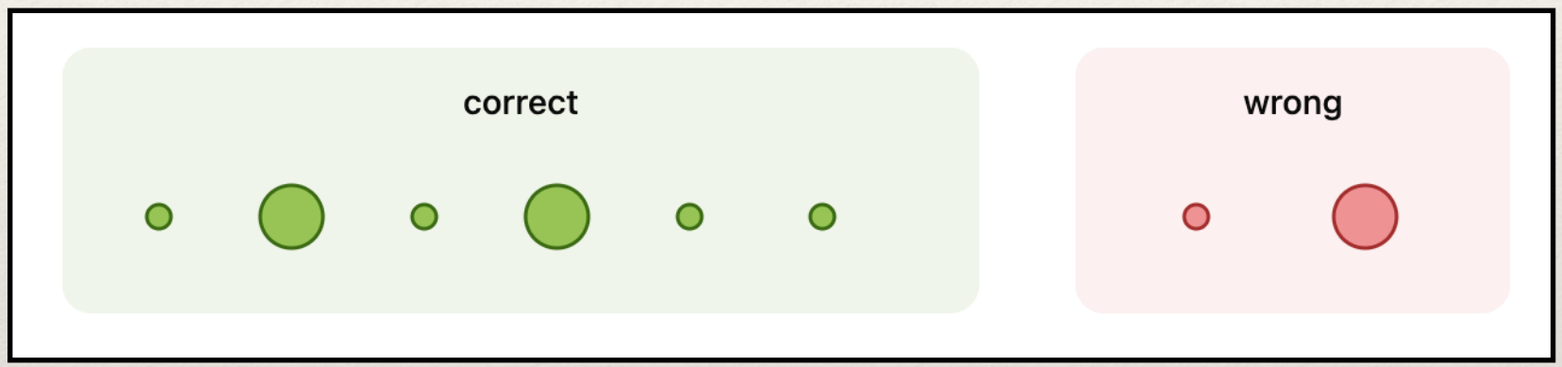
$$\begin{cases} w_i^{(m)} = w_i^{(m-1)} \sqrt{\frac{\epsilon_{m-1}}{1-\epsilon_{m-1}}} & \text{if} \quad h_{m-1}(x_i) = y_i, \\ w_i^{(m)} = w_i^{(m-1)} \sqrt{\frac{1-\epsilon_{m-1}}{\epsilon_{m-1}}} & \text{if} \quad h_{m-1}(x_i) \neq y_i, \end{cases} \qquad i = 1, \ldots, n,$$

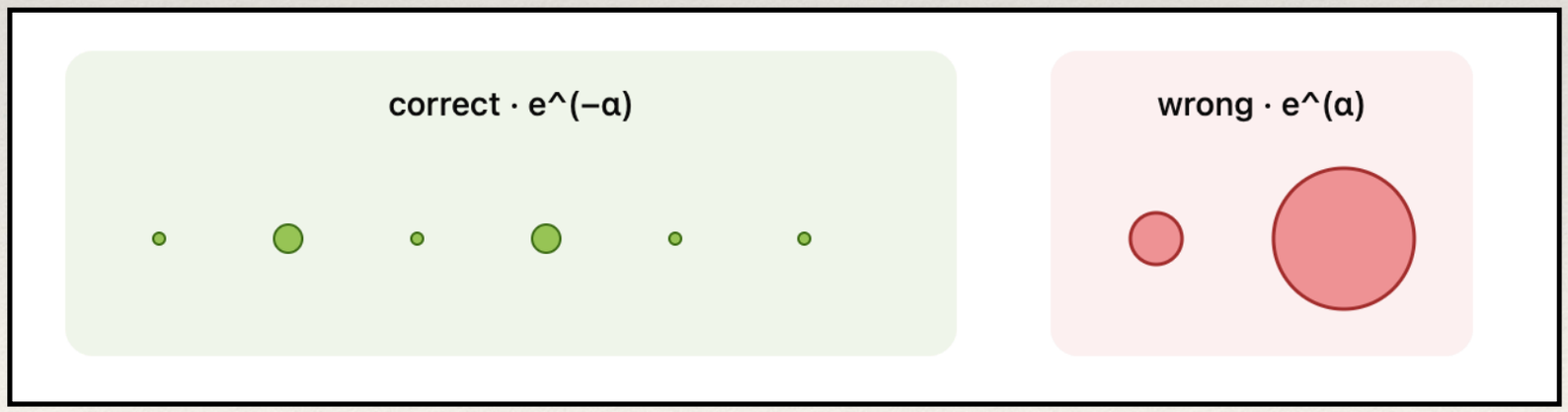
and provide an interpretation of these weight updates.

Q1g Analyze

Loss(Ensemble) = ∑ size * Loss(new model_m)

$$e^{-\alpha_m} \sum_{i:h_m(x_i)=y_i} w_i^{(m)} + e^{\alpha_m} \sum_{i:h_m(x_i)\neq y_i} w_i^{(m)}.$$



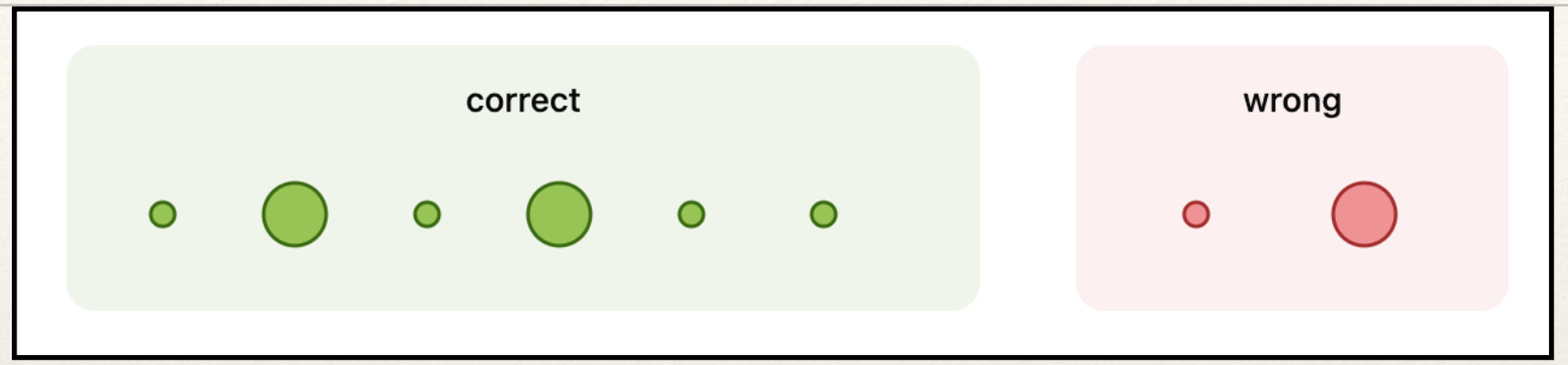


$e^{-\alpha_m}$

Correct: Scaling down Wrong: Scaling up

e^{α_m}

Q1g Analyze



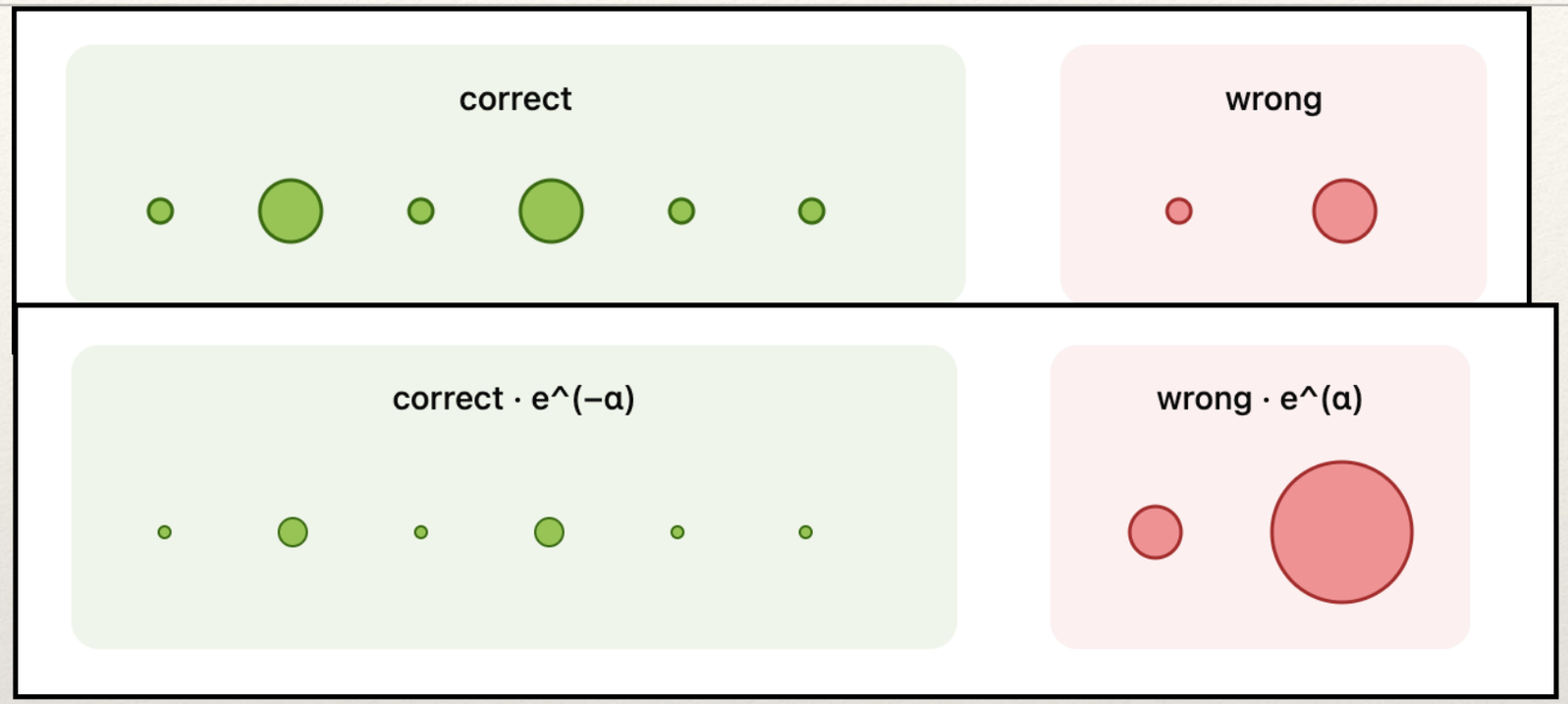
$e^{-\alpha_m}$

Correct: Scaling down Wrong: Scaling up

e^{α_m}

$$\alpha_m^* = \frac{1}{2} \ln \left(\frac{1 - \epsilon_m}{\epsilon_m} \right).$$

Q1g Answer



$e^{-\alpha_m}$

Correct: Scaling down Wrong: Scaling up

e^{α_m}

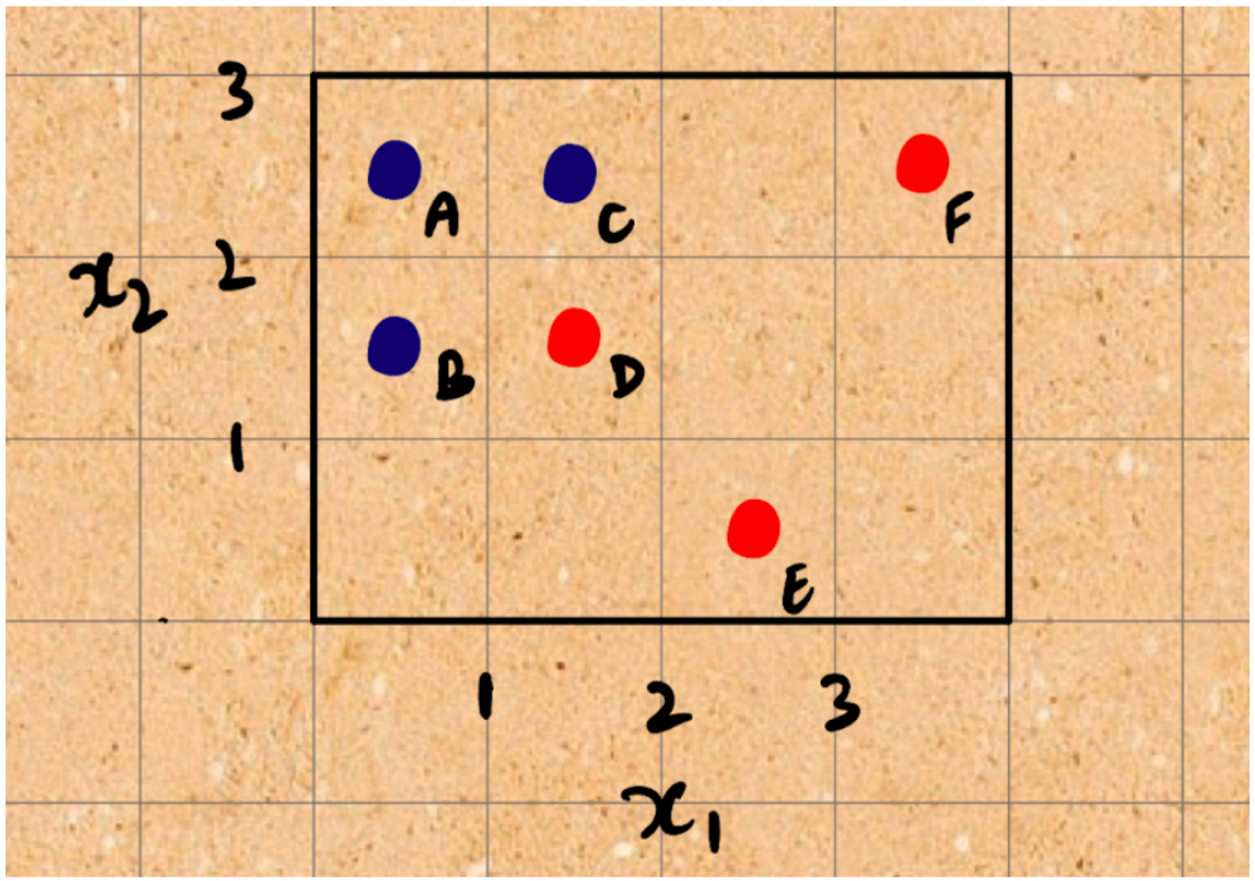
$$\alpha_m^* = \frac{1}{2} \ln \left(\frac{1 - \epsilon_m}{\epsilon_m} \right).$$

$$\begin{cases} w_i^{(m)} = w_i^{(m-1)} \sqrt{\frac{\epsilon_{m-1}}{1-\epsilon_{m-1}}} & \text{if} \quad h_{m-1}(x_i) = y_i, \\ w_i^{(m)} = w_i^{(m-1)} \sqrt{\frac{1-\epsilon_{m-1}}{\epsilon_{m-1}}} & \text{if} \quad h_{m-1}(x_i) \neq y_i, \end{cases} \qquad i = 1, \ldots, n,$$

Question 2

Question 2. AdaBoost Toy Example

We will now implement AdaBoost by hand on a toy example. Consider the following two-dimensional points labelled A, B, C, D, E, F. The blue points have true label -1 and the red points have true label $+1$. The size of each of the points is equal since we are starting out with uniform weights, $w_i^{(1)} = \frac{1}{6}$ for $i = 1, \ldots, 6$.



Red: +1

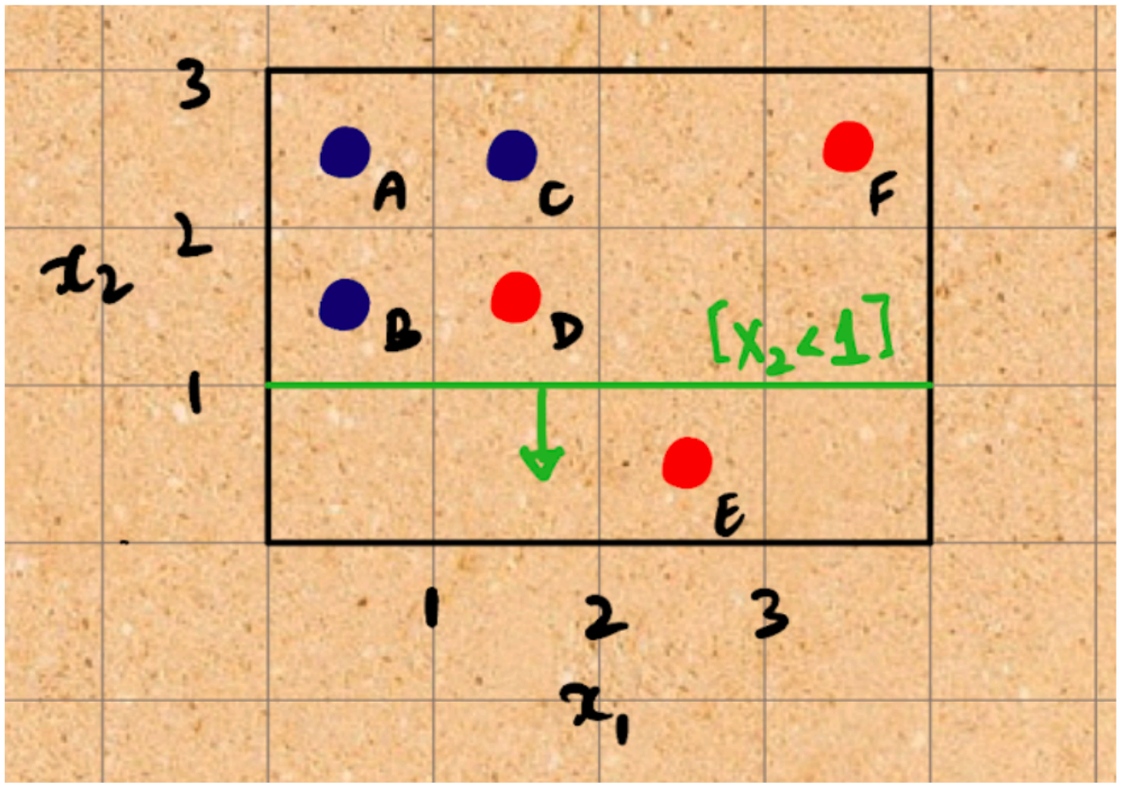
Blue: -1

Question 2

We consider the following class of simple hypotheses:

$$\mathcal{H} = \{[x_1 > 1], [x_1 > 2], [x_1 > 3], [x_2 < 1], [x_2 < 2]\}.$$

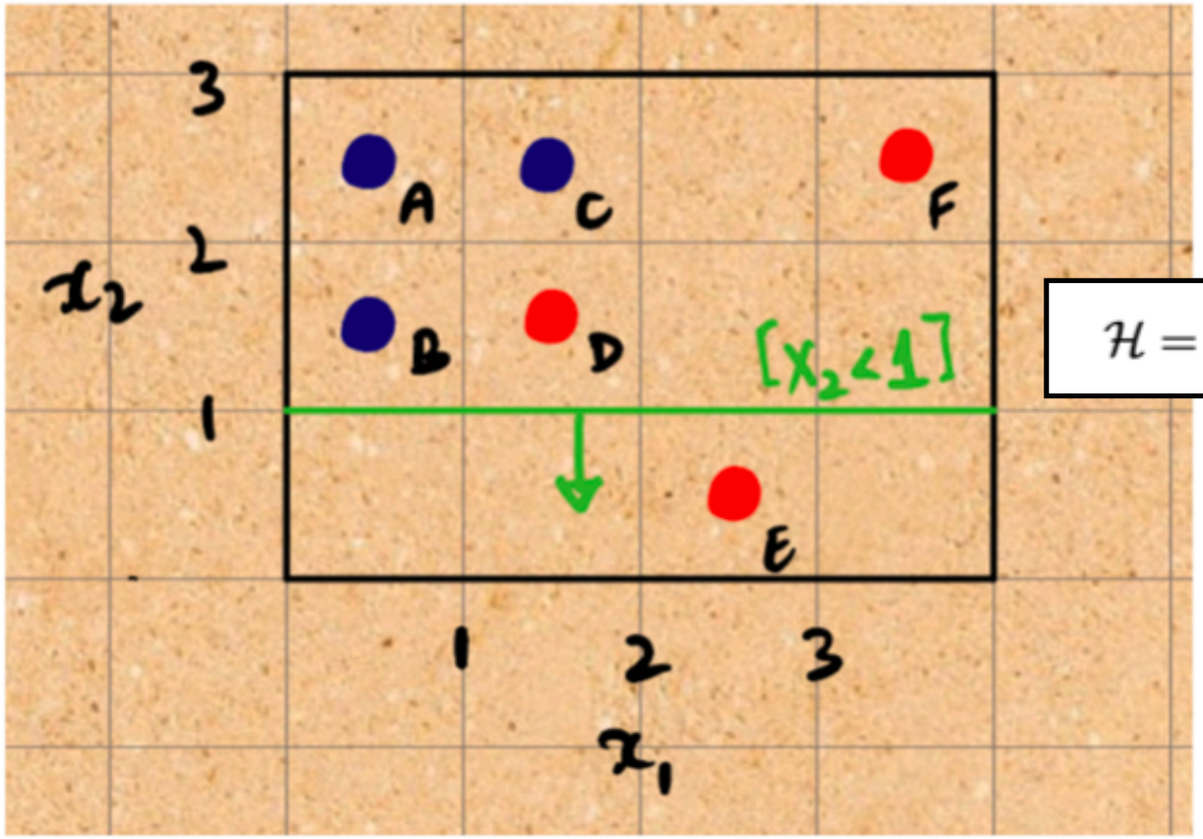
In the following plot, we see an example of what the hypothesis $[x_2 < 1]$ looks like. It classifies all points with feature value x_2 smaller than 1 as a positive point. Note that the small arrow indicates that this classifier labels all points in the direction of this arrow as belonging to the class '+1'. In this plot, the hypothesis classifies all points correctly except for points D and F.



Now that we have our set of hypothesis (base learners), we can proceed with implementing AdaBoost. Note throughout, we will implement the normalized version of AdaBoost, and you should round up to two decimal places for simplicity.

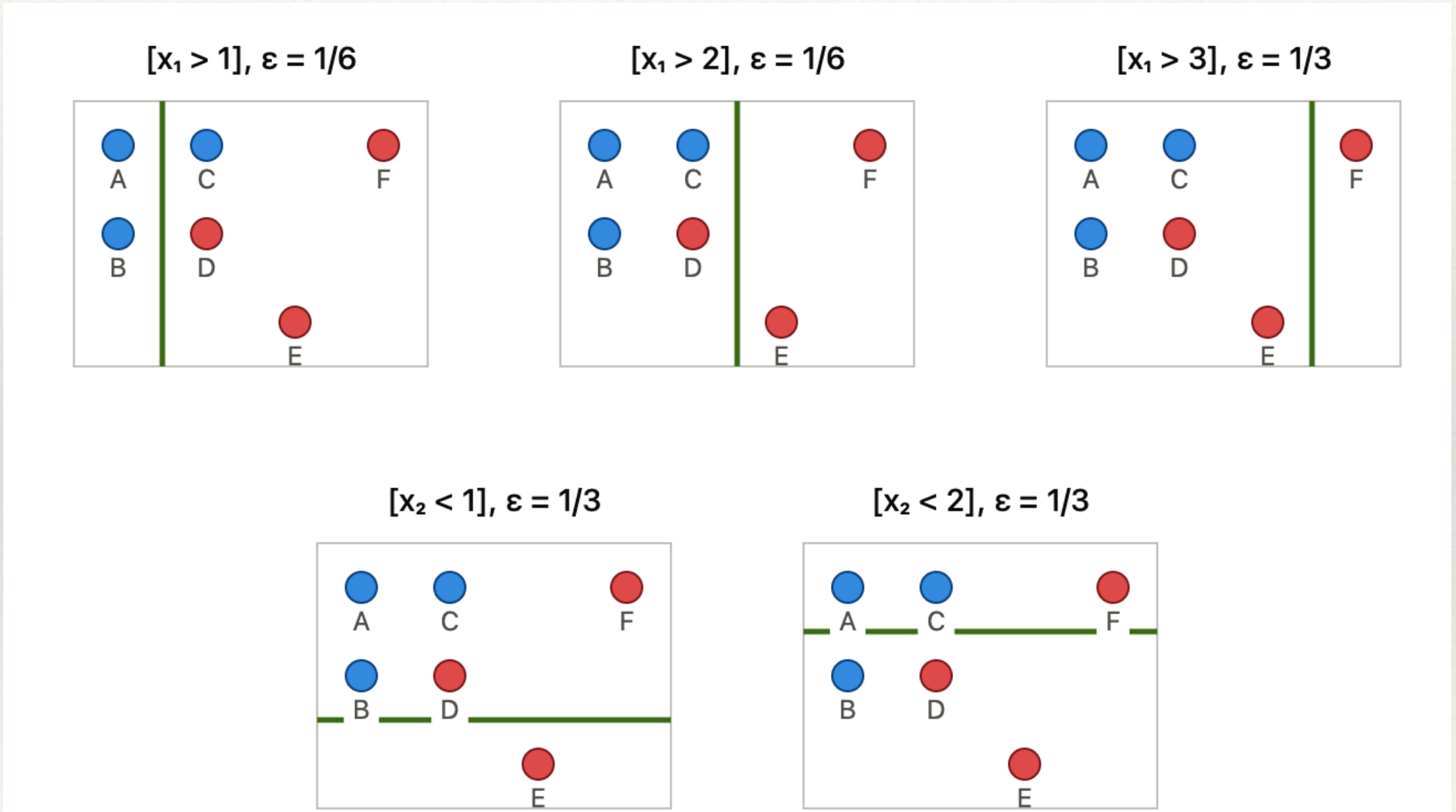
Q2a

(a) Find h_1^* that minimizes the weighted misclassification error $\epsilon_1 = \sum_{y_i \neq h_1(x_i)} w_i^{(1)}$. To do this, create a table with two columns titled 'hypothesis' and 'ϵ_1' and one row for each hyothesis. Further, find the corresponding weight α_1. Also, find the new weights $w_i^{(2)}$ Plot your Adaboost classifier C_1 and the data (with new weights). Note that if any two hypotheses are tied, pick the one that is listed first in the definition of $\mathcal{H}$.

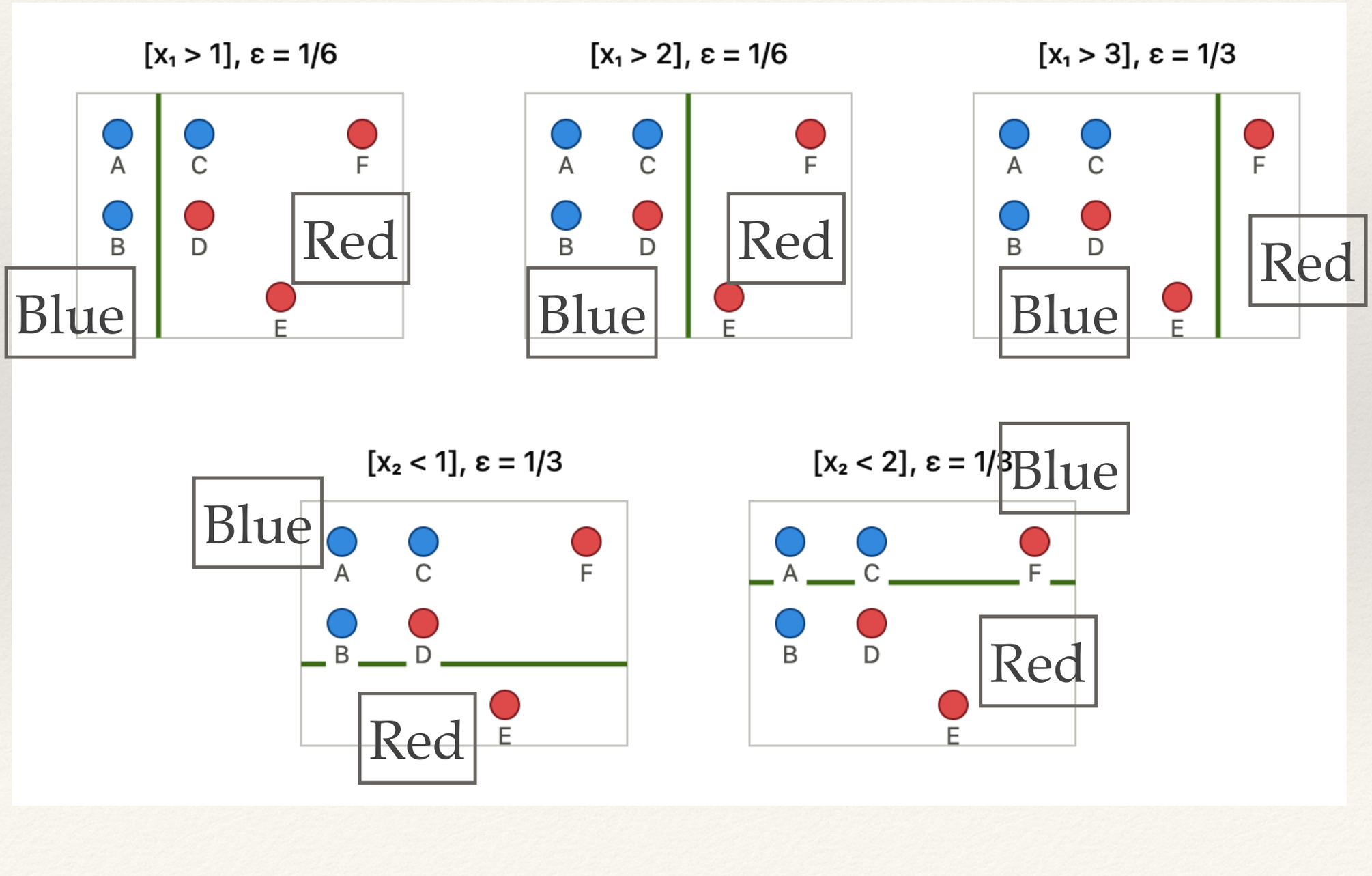


$$\mathcal{H} = \{[x_1 > 1], [x_1 > 2], [x_1 > 3], [x_2 < 1], [x_2 < 2]\}.$$

Q2a Analyze: Error of hypothesis



Q2a Analyze: Error of hypothesis



Q2a Analyze: alpha

Note that if any two hypotheses are tied, pick the one that is listed first in the definition of $\mathcal{H}$.

Hypothesis	ϵ_1
$[x_1 > 1]$	$1/6$
$[x_1 > 2]$	$1/6$
$[x_1 > 3]$	$1/3$
$[x_2 < 1]$	$1/3$
$[x_2 < 2]$	$1/3$

From Q1f:

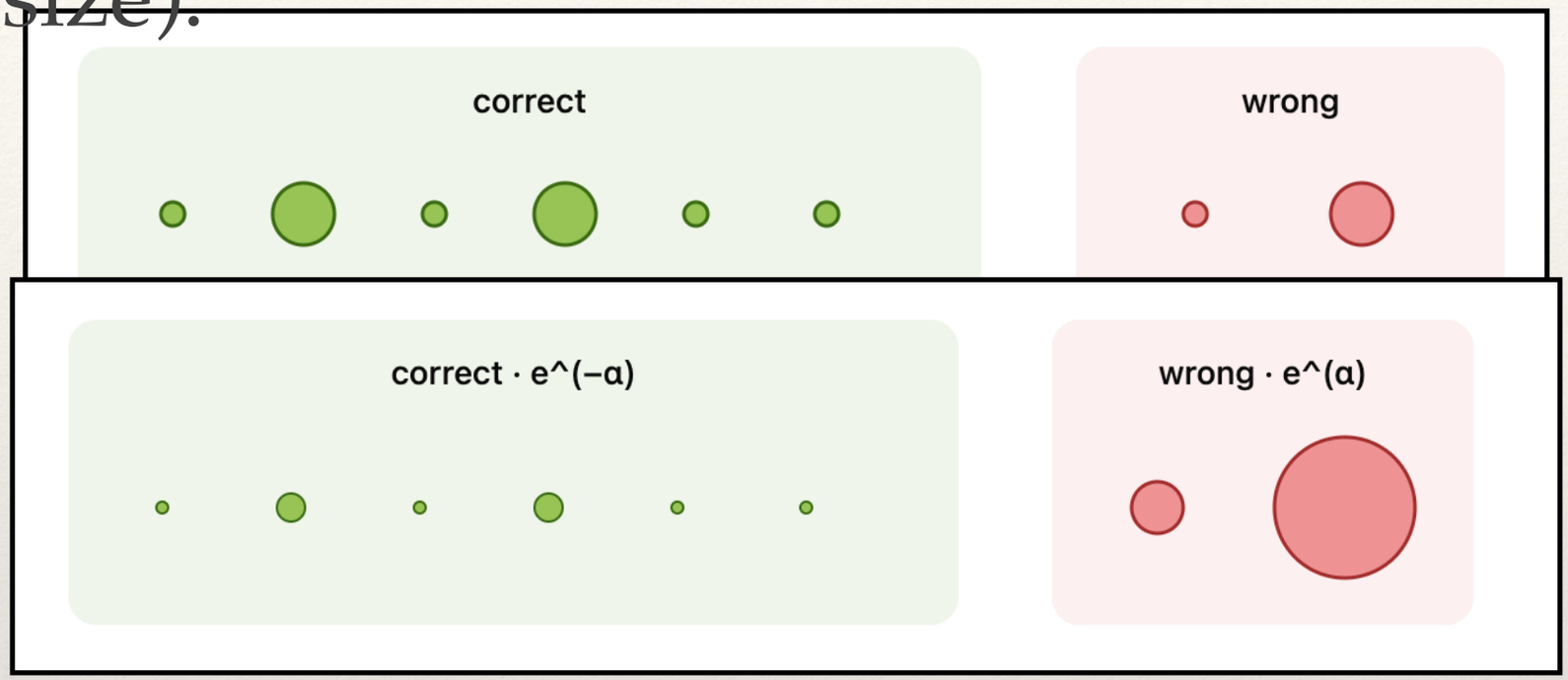
$$\alpha_m^* = \frac{1}{2} \ln \left(\frac{1 - \epsilon_m}{\epsilon_m} \right).$$

$$\alpha_1^* = \tfrac{1}{2} \ln \left(\frac{1 - 1/6}{1/6} \right) = \tfrac{1}{2} \ln \left(\frac{5/6}{1/6} \right) = \tfrac{1}{2} \ln 5$$

Q2a Analyze: weight update

Initial weight(~~size~~):

1/6



$e^{-\alpha_m}$

Correct: Scaling down

Wrong: Scaling up

e^{α_m}

$$e^{-\alpha_1} = 1/\sqrt{5} \approx 0.447$$

$$e^{\alpha_1} = e^{\frac{1}{2}\ln 5} = \sqrt{5} \approx 2.236$$

1/6 * 0.447 =0.0745

1/6 * 2.236 =0.373

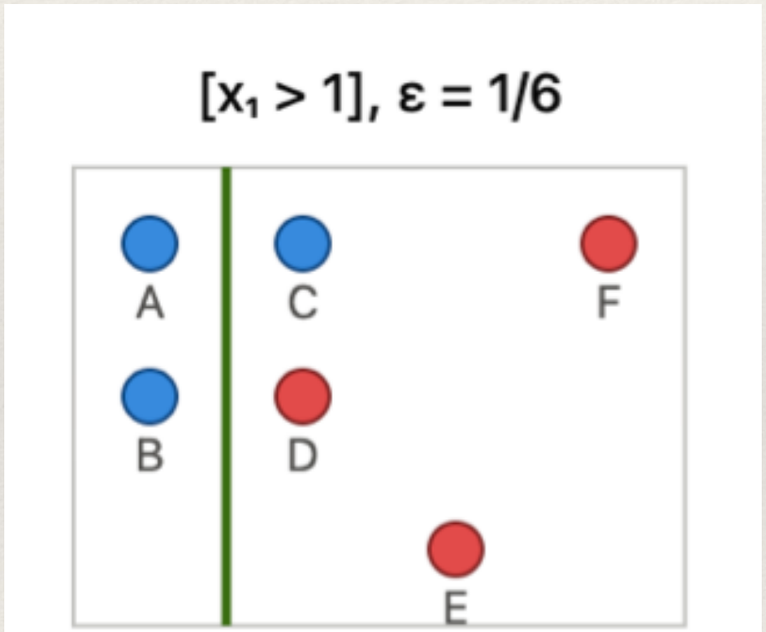
Q2a Analyze: weight update

Correct: Scaling down

$1/6 * 0.447 = 0.0745$

Wrong: Scaling up

$1/6 * 2.236 = 0.373$



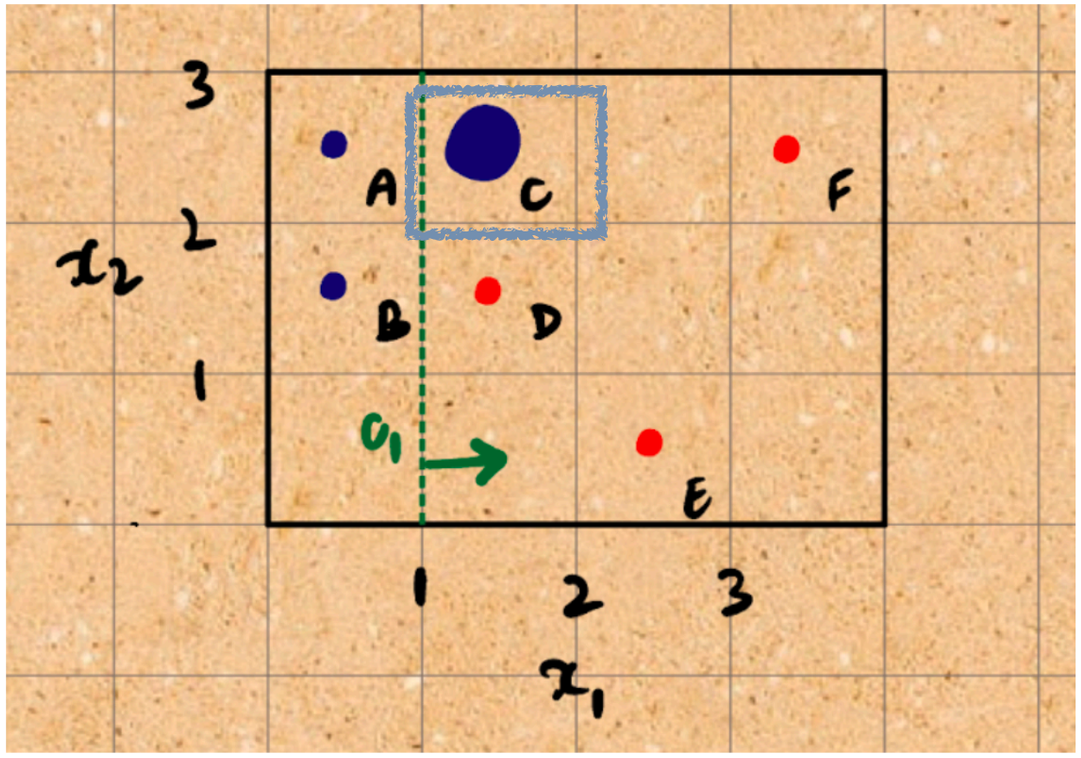
Point	$w^{(1)}$	Factor	$w^{(2)}$
A	1/6	× 0.447	0.0745
B	1/6	× 0.447	0.0745
C	1/6	× 2.236	0.373
D	1/6	× 0.447	0.0745
E	1/6	× 0.447	0.0745
F	1/6	× 0.447	0.0745

Q2a Analyze: Normalization

Point	$w^{(1)}$	Factor	$w^{(2)}$
A	1/6	× 0.447	0.0745
B	1/6	× 0.447	0.0745
C	1/6	× 2.236	0.373
D	1/6	× 0.447	0.0745
E	1/6	× 0.447	0.0745
F	1/6	× 0.447	0.0745

Q2a Analyze: Normalization

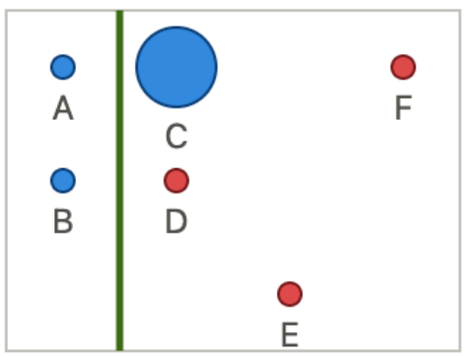
Point	$w^{(1)}$	Factor	$w^{(2)}$	Normalised $w^{(2)}$
A	1/6	× 0.447	0.0745	0.1
B	1/6	× 0.447	0.0745	0.1
C	1/6	× 2.236	0.373	0.5
D	1/6	× 0.447	0.0745	0.1
E	1/6	× 0.447	0.0745	0.1
F	1/6	× 0.447	0.0745	0.1



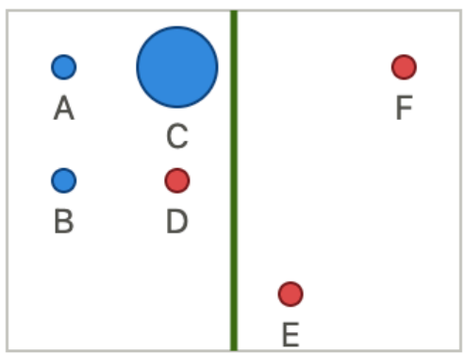
Q2b Analyze

(b) Perform one more round of AdaBoost, report the same quantities and provide a plot of C_2. To create the plot, figure out what C_2 predicts on each of the 6 points. Then draw the classifier accordingly.

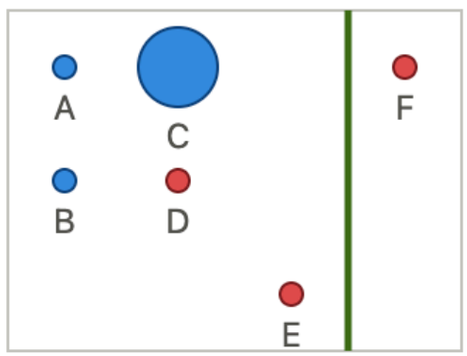
$[x_1 > 1]$, $\varepsilon_2 = 0.5$



$[x_1 > 2]$, $\varepsilon_2 = 0.1$

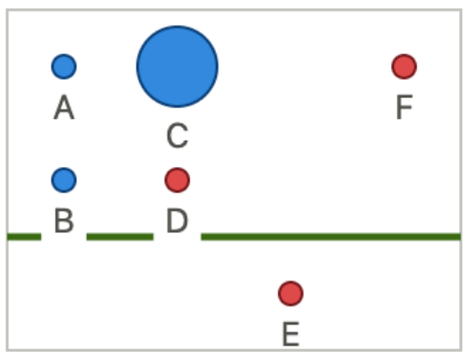


$[x_1 > 3]$, $\varepsilon_2 = 0.2$

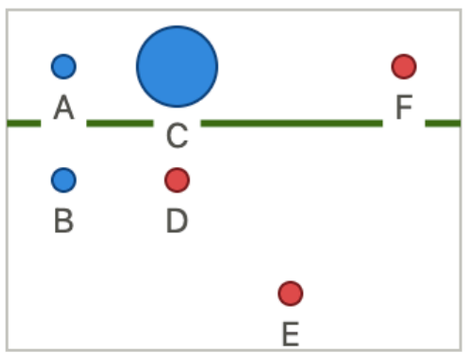


Normalised $w^{(2)}$
0.1
0.1
0.5
0.1
0.1
0.1

$[x_2 < 1]$, $\varepsilon_2 = 0.2$



$[x_2 < 2]$, $\varepsilon_2 = 0.6$



Q2b Analyze

Hypothesis	ϵ_2
$[x_1 > 1]$	$1/2$
$[x_1 > 2]$	$1/10$
$[x_1 > 3]$	$1/5$
$[x_2 < 1]$	$1/5$
$[x_2 < 2]$	$1/5$

From Q1f:

$$\alpha_m^* = \frac{1}{2} \ln \left(\frac{1 - \epsilon_m}{\epsilon_m} \right).$$

$$\alpha_2^* = \tfrac{1}{2} \ln \left(\frac{1 - 1/10}{1/10} \right) = \tfrac{1}{2} \ln 9 = \ln 3 \approx 1.099$$

Q2b Analyze

Point	$w^{(2)}$	Factor	$w^{(3)}$	Normalised $w^{(3)}$
A	0.1	× 1/3	0.0333	1/18 ≈ 0.056
B	0.1	× 1/3	0.0333	1/18 ≈ 0.056
C	0.5	× 1/3	0.1667	5/18 ≈ 0.278
D	0.1	× 3	0.3	1/2 = 0.5
E	0.1	× 1/3	0.0333	1/18 ≈ 0.056
F	0.1	× 1/3	0.0333	1/18 ≈ 0.056

Q2b Analyze: Ensemble Model

$$\alpha_1^* = \tfrac{1}{2}\ln\left(\frac{1-1/6}{1/6}\right) = \tfrac{1}{2}\ln\left(\frac{5/6}{1/6}\right) = \tfrac{1}{2}\ln 5$$

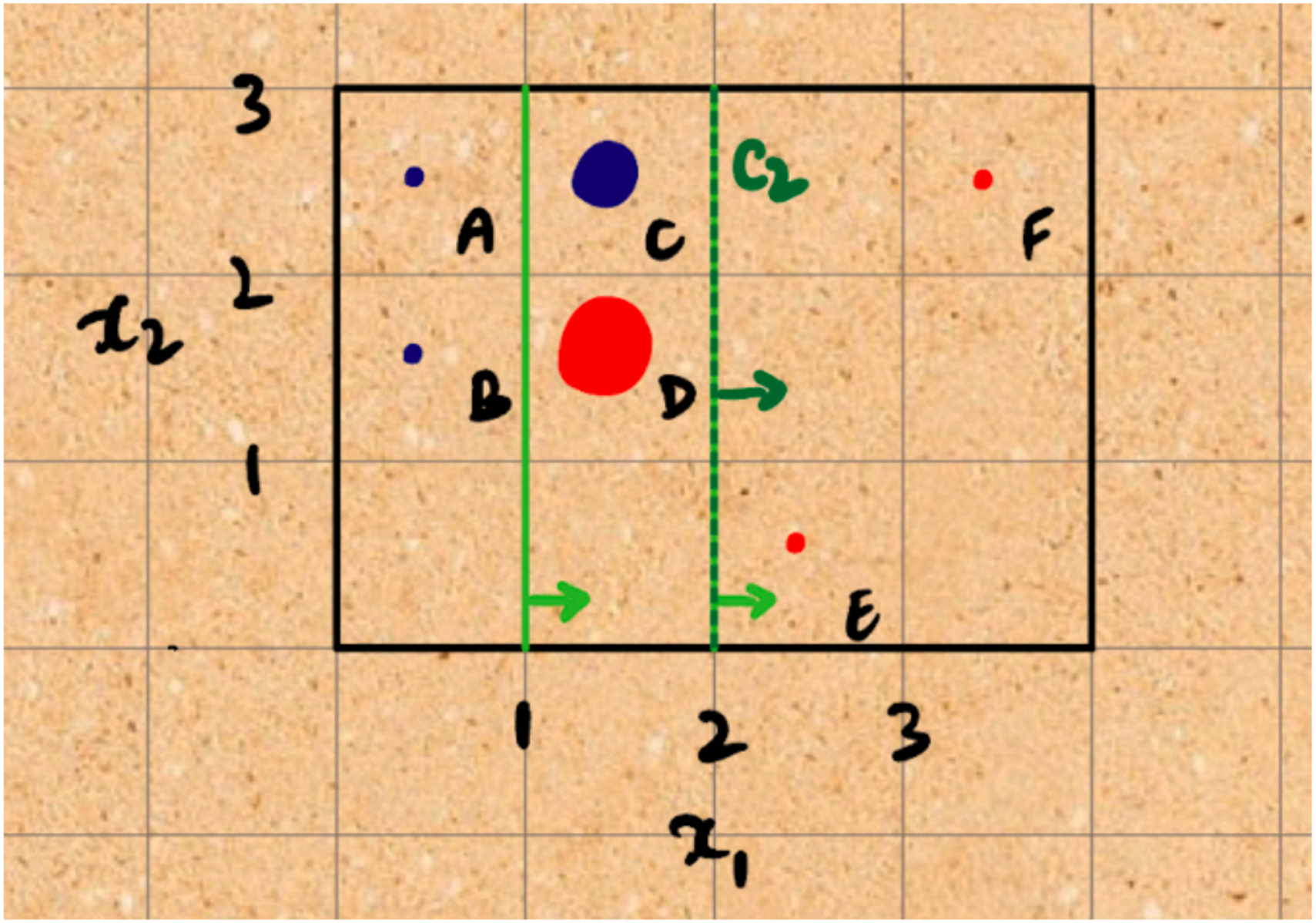
$$\alpha_2^* = \tfrac{1}{2}\ln\left(\frac{1-1/10}{1/10}\right) = \tfrac{1}{2}\ln 9 = \ln 3 \approx 1.099$$

$$C_2 = 0.8[x_1 > 1] + 1.1[x_1 > 2].$$

Q2b Analyze: Ensemble Prediction

Point	x_1	$h_1 = [x_1 > 1]$	$h_2 = [x_1 > 2]$	$C_2 = 0.8\,h_1 + 1.1\,h_2$	$\text{sign}(C_2)$	y	Result
A	0.5	−1	−1	−0.8 − 1.1 = −1.9	−1	−1	✓
B	0.5	−1	−1	−0.8 − 1.1 = −1.9	−1	−1	✓
C	1.5	+1	−1	0.8 − 1.1 = −0.3	−1	−1	✓
D	1.5	+1	−1	0.8 − 1.1 = −0.3	−1	+1	✗
E	2.5	+1	+1	0.8 + 1.1 = 1.9	+1	+1	✓
F	3.5	+1	+1	0.8 + 1.1 = 1.9	+1	+1	✓

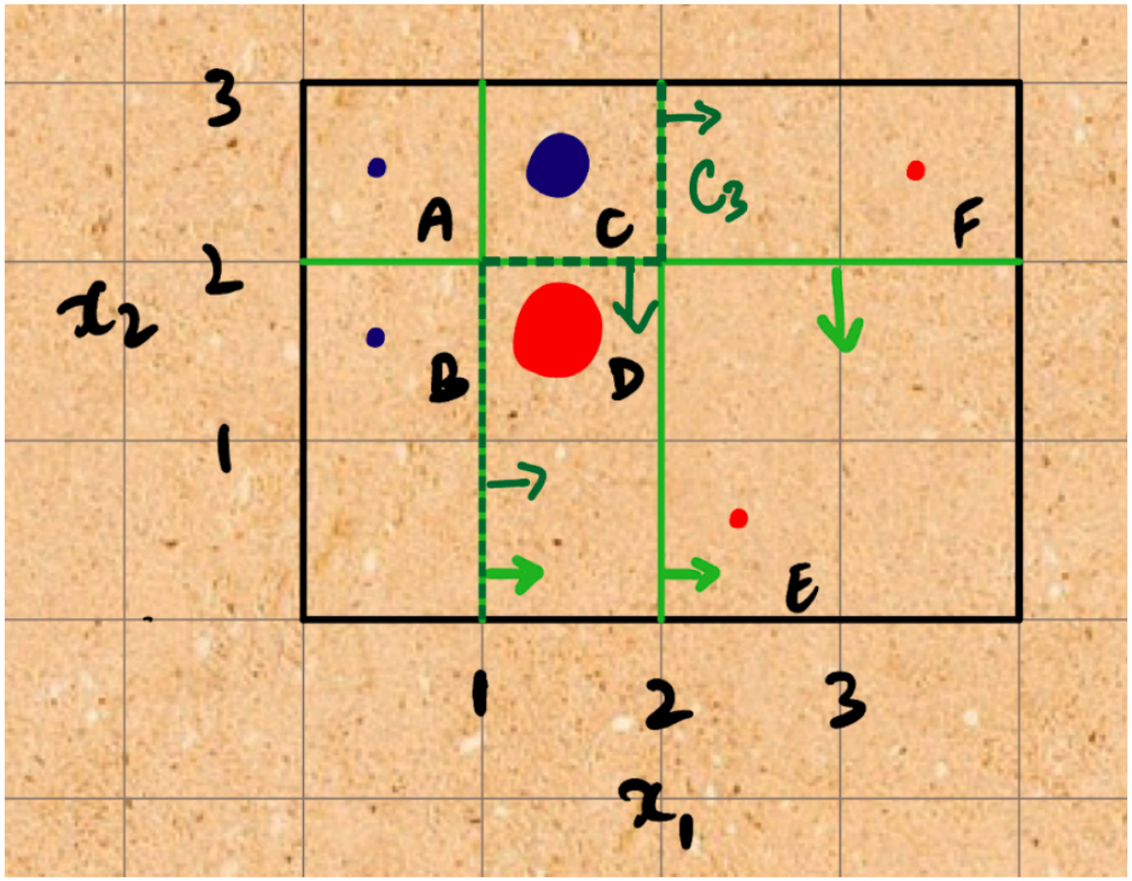
Q2b Plot



Q2c/Q2d

(c) Perform one final round of AdaBoost, report the same quantities and provide a plot of C_3, except that you do not need to re-calculate the weights this time.

(d) Comment on the plot of C_3. What do you notice?



Thank you!